



The Risk-Bearing Subject of AI-Related Existential Risk to Humanity

From Species Survival to the Realization of Subjecthood

under Conditions of Human Civilization

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Abstract

This paper is a work of conceptual engineering on the risk-bearing subject of AI-related existential risk to humanity. It also has significance for social epistemology, philosophy of technology, and AI risk governance. Existing discussions usually understand “humanity” as the biological species *Homo sapiens*, and accordingly define the risk in terms of species extinction, permanent incapacitation, or future deprivation. This paper preserves the baseline status of that framework, but proposes a paradigm extension: while the risk-bearing subject remains humanity as an embodied species, the possible modes of existential harm to that subject should be extended from destruction or permanent incapacitation at the single level of biological survival to the irreversible or difficult-to-reverse weakening of the foundational conditions under which humanity realizes subjecthood within civilization.

The central question of this paper is: when we say that AI may threaten “humanity,” what exactly is the “humanity” at risk? The paper distinguishes two basic layers of possible harm

to humanity as the risk-bearing subject: the biological-survival layer and the civilizational-conditions layer. The former concerns the survival of humanity as an embodied species; the latter concerns humanity's capacity, within an intergenerational common world, to know what is true, form judgment, bear responsibility, participate in institutions, and create meaning. Within the civilizational-conditions layer, the paper further identifies the category-calibration mechanism: the mechanism through which foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning are clarified, maintained, and revised. The philosophical kernel is not an independent entity, but the category-calibration mechanism internal to civilizational conditions that maintains the operability of foundational categories.

AI may weaken humans' substantive participation in the functional systems of civilization, or destabilize the calibration of foundational categories, through **deprivation by substitution** or **mediated reconfiguration**. The expanded definition proposed in this paper is as follows: AI-related existential risk to humanity refers to the risk, arising from the development, deployment, integration, or loss of control of AI systems, that AI may irreversibly destroy the human species; or render humanity permanently incapacitated, or otherwise permanently foreclose humanity's future; or undermine—irreversibly or in ways that are difficult to reverse—the foundational conditions under which humanity, as an embodied species, realizes its subjecthood within civilization. Within this definition, risks to the realization of human subjecthood at the civilizational-conditions layer are constrained by **four thresholds**: they must act upon foundational civilizational conditions, possess cross-domain diffusion capacity, be capable of causing irreversible or difficult-to-reverse structural damage, and cause the first three forms of damage to converge into overall damage to humanity's capacity as a future-oriented subject.

What this paper provides is not a complete risk taxonomy, nor a technical control scheme or policy checklist, but an **ontological coordinate system** needed at the level of risk identification in AI safety, AI risk analysis, and AI governance. Before discussing specific risk types, governance pathways, or technical solutions, we must first clarify the risk-bearing subject called "humanity" and the possible modes of harm to it.

Keywords

artificial intelligence; AI safety; AI risk; AI governance; existential risk to humanity; risk-bearing subject; conceptual engineering; social epistemology; philosophy of technology; human civilization; civilizational conditions; human subjecthood; functional systems of civilization; philosophical kernel; foundational categories; category calibration; realization of subjecthood

1 | Introduction: AI Reopens the Question of the Risk-Bearing Subject of Existential Risk to Humanity

1.1 The Problem: Technological Background and Philosophical Question

Artificial intelligence is reopening the question of existential risk to humanity. The risks discussed in this paper are situated against the background of the **AGI era** [Note 1]. By “AGI era,” this paper does not mean that some recognized threshold for AGI has already been crossed. Rather, it refers to an ongoing historical transition in which AI systems are becoming more general, more autonomous, more capable of action, and more deeply embedded in institutions. Many of the risk mechanisms discussed here can already be observed in the early stages of this transition—in current applications of advanced AI, generative AI, and agentic AI.

What is distinctive about current AI is that its expansion of capabilities is entering the foundational structures through which humans know the world, form judgment, organize action, allocate responsibility, govern institutions, and understand themselves. AI is no longer merely a tool that extends external capacities. It is becoming a mediator of human knowledge, institutions, action, and meaning. [Note 2] It generates code, organizes evidence, drafts reports, assists research, structures workflows, recommends decisions, and participates in institutional processes in business, education, scientific research, law, medicine, administration, and public services. It is also beginning to participate in the development of AI systems themselves, including assisted coding, model evaluation, research automation, and technological iteration.

This is why current AI risk is not only a technical problem, but also a philosophical one. The technical problem asks whether systems are reliable, safe, controllable, and compliant. The philosophical problem asks: what is truth, who is the subject, what is responsibility, what has value, and why do humans still need to judge and act for themselves? When AI mediates knowledge, action, institutions, judgment, and meaning, it affects not only what humans do, but also how humans understand themselves, how they bear responsibility, and how they organize a common world.

1.2 Conceptual Slippage in the Risk-Bearing Subject: Implicit Definitions of “Humanity” across Domains

Current discussions of AI risk implicitly rely on different risk-bearing subjects, but rarely first clarify what “humanity” is. Alignment research usually presupposes that what must be protected is **human preference**; biosecurity research presupposes **human life**; governance research presupposes **human social order**; philosophical discussion presupposes **human value and dignity**; and the classical framework of existential risk focuses primarily on whether **the human species** (*Homo sapiens*) might become extinct. These concepts of “humanity” are not mutually compatible. When they are mixed together in the same policy document or technical roadmap, risk identification continually slides between dimensions: at one moment it discusses an epistemic problem, such as whether a model outputs truth;

at the next it shifts to biosecurity, such as whether a model can be misused; and then it becomes an economic-adaptation problem, such as whether employment is being replaced. If the risk-bearing subject has not yet been clarified, different fields of inquiry will never share a common set of ontological coordinates.

Many discussions of AI existential risk still presuppose a relatively narrow object of risk. They focus mainly on whether AI might destroy the human species, seize critical infrastructure, permanently deprive humanity of its future, or decisively deprive humans of control. These risks are certainly serious. Species extinction, permanent incapacitation, and irreversible catastrophe are the most basic forms of existential risk. This paper does not attempt to weaken their importance. [Note 3]

Traditional frameworks often presuppose an extreme scenario: an AI system may develop an autonomous objective function and enter into decisive conflict with human interests. This paper does not deny the seriousness of that scenario, but it also does not depend on it. The risk mechanisms discussed here may already arise when AI has not formed autonomous goals and is functioning only as a deep mediator of human knowledge, institutions, and meaning.

1.3 Core Claim and Methodological Position

This paper argues that the object of AI-related existential risk to humanity may extend beyond this narrower framework. When we say that AI may threaten “humanity,” does “humanity” refer only to the continuation of *Homo sapiens* as a biological species? Or should humanity also be understood as an embodied species that, under civilizational conditions, realizes the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future?

If humanity is understood only as a biological species, then AI-related existential risk to humanity is mainly confined to physical destruction, species extinction, permanent incapacitation, or loss of control over humanity's future. But if humanity is also understood as an embodied species that realizes subjecthood under civilizational conditions, then AI may produce another form of existential harm: the human species continues to exist, institutions and technological systems continue to operate, and society continues to produce, consume, and communicate, yet humans gradually lose the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future. In such a condition, humans have not become biologically extinct, yet humanity may be deeply weakened as a future-oriented subject.

The core claim of this paper is that the risk-bearing subject of AI-related existential risk to humanity always remains humanity as an embodied species. This paper does not expand the risk-bearing subject to include civilization, institutions, the philosophical kernel, or AI systems. What it expands is the range of possible modes of existential harm to humanity as that subject: from destruction, permanent incapacitation, or future deprivation at the single biological level to the irreversible or difficult-to-reverse weakening of the foundational conditions under which humanity realizes its subjecthood within civilization.

To this end, the paper undertakes a work of conceptual engineering [Note 4]. It distinguishes two basic layers of possible harm to humanity as the risk-bearing subject, and separately identifies the category-calibration mechanism within the civilizational-conditions layer. The biological-survival layer explains the survival of humanity as an embodied biological population. The civilizational-conditions layer explains how humans realize subjecthood through language, knowledge, institutions, structures of responsibility, functional systems of civilization, and narratives of meaning. The category-calibration mechanism explains how foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning are clarified, maintained, and revised within civilizational conditions. The philosophical kernel is not a third parallel subject, nor an independent entity, but the category-calibration mechanism internal to civilizational conditions that maintains the operability of foundational categories.

This paper therefore provides neither a complete risk taxonomy nor a technical control scheme or policy checklist. It provides an **ontological coordinate system at the level of risk identification**: before discussing specific risk types, governance pathways, or technical solutions, we must first clarify the risk-bearing subject called “humanity” and the possible modes of harm to it. Only within this coordinate system can technical research, policy discussion, philosophical analysis, and social observation be placed within a framework that is comparable, connectable, and open to critique.

1.4 Structure of the Argument

The paper proceeds as follows. Section 2 offers a two-dimensional working definition of humanity as an embodied species, and explains why understanding humanity only at the biological-survival layer is necessary but insufficient. Section 3 defines human civilization as the intergenerational condition for the realization of subjecthood. Section 4 argues that the philosophical kernel is not an independent entity within civilization, but a category-calibration mechanism internal to civilizational conditions. Section 5 uses a functional analogy to help readers understand the relations among biological substrate, category interfaces, civilizational operating environment, and functional systems of civilization. Section 6 explains why AI makes these conditions matters of existential risk, and proposes the expanded definition and four thresholds. Section 7 concludes by explaining the significance of this conceptual reconstruction for AI safety, AI risk analysis, and research on humanity’s future.

2 | Humanity as an Embodied Species: A Two-Dimensional Working Definition

If “humanity” is understood only as *Homo sapiens*, a biological species, then existential risk will primarily point to species extinction, permanent incapacitation, and future deprivation. This understanding captures the most basic form of risk, but it is not sufficient to exhaust the object of human risk in the AI era. The task of this section is to provide a working defi-

inition of “humanity” and to explain why humanity, as an embodied species, constitutes the baseline dimension of the risk-bearing subject and yet cannot be reduced to the biological-survival layer alone.

2.1 The Two-Dimensional Working Definition of Humanity

In this paper, “humanity” first refers to *Homo sapiens* as an embodied biological population. Humans have bodily structures, nervous systems, capacities for perception, cognitive capacities, reproductive mechanisms, physiological vulnerabilities, mortality, and dependence on ecological and material environments. This biological-survival layer constitutes the physical precondition of human existence. If the human species becomes extinct, or if humanity permanently loses the capacities to perpetuate itself, to act, and to shape the future, then all discussions of human civilization, institutions, responsibility, value, and meaning lose their bearer.

Humanity, however, does not exist only at the biological-survival layer. Humanity as an embodied species also realizes its subjecthood under civilizational conditions: through language, knowledge, education, institutions, structures of responsibility, public reason, historical memory, and narratives of meaning, humans acquire and sustain the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future.

Accordingly, the working definition of “humanity” in this paper has two layers. First, humanity is an embodied biological species. Second, humanity is an embodied species that realizes subjecthood under civilizational conditions. The former identifies the biological substrate of human existence; the latter explains why humanity is not merely a biological population, but a historical form of existence capable of knowing, judging, taking responsibility, transmitting and creating meaning, and acting toward the future.

This two-dimensional working definition does not change the risk-bearing subject. The risk-bearing subject is always humanity as an embodied species. Civilization, institutions, cultures, technological systems, and the philosophical kernel are not risk-bearing subjects parallel to humanity. What this paper expands is not the risk-bearing subject itself, but the possible modes of existential harm to that subject. Within this framework, the possible modes of harm to humanity mainly include the biological-survival layer and the civilizational-conditions layer; the category-calibration mechanism is a mechanism within the civilizational-conditions layer that must be separately identified. It is responsible for the clarification, maintenance, and revision of foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. This will be developed in Section 4.

2.2 Species Extinction as the Baseline Risk

Human species extinction is the most direct and stringent form of existential risk to humanity. If *Homo sapiens* no longer exists in the biological sense, or if humanity permanently loses the capacities to reproduce, act, and shape the future, then all possibilities for human

civilization and humanity's future are closed. Without embodied human life, there can be no continued unfolding of human civilization and no bearer of the realization of human subjecthood.

Species-level risk is the baseline that any expanded definition must preserve. Without the continuation of humanity as embodied life, civilizational conditions and the realization of subjecthood cannot arise. The point of this paper is only that species survival is a necessary condition of humanity's future, but not a sufficient condition. Humanity may continue to exist biologically while gradually losing, in civilizational terms, certain foundational capacities that enable it to continue existing as a knower, judge, bearer of responsibility, institutional participant, and creator of meaning.

2.3 Embodiment: Humanity Is Not a Pure Information Structure

To emphasize humanity as an embodied species is to reject the reduction of humanity to a pure information structure, an abstract rational subject, or a computational pattern that can be arbitrarily migrated. Human cognition, emotion, judgment, learning, relationships, responsibility, and meaning are all closely tied to bodily conditions. Perceiving the world depends on the body. Language learning depends on embodied experience and social interaction. Bearing responsibility depends on the capacity to act and to endure the consequences of action. Meaning-formation is also bound up with birth, death, time, suffering, dependence, relations, and finitude.

This point must be preserved especially in the context of AI. As AI systems increasingly perform cognitive tasks, generate text, organize knowledge, propose solutions, and participate in decision-making, humans are easily understood as cognitive entities replaceable by more efficient information-processing systems. Human existence, however, is not merely information processing. Truth, responsibility, subjecthood, and meaning in human civilization are all connected to embodied finitude. A being that can die, be injured, make commitments, feel regret, grow in time, and bear consequences does not occupy the same existential structure as an information system without body, death, or vulnerability.

Emphasizing embodiment does not mean that human civilization excludes nonhuman technological components. Writing, tools, machines, cities, platform systems, and AI systems can all become components or mediating structures of human civilization. The key point is that even when technology deeply enters civilization, the risk-bearing subject in this paper remains embodied humanity, not technological systems, civilizational systems, or AI systems themselves. Technology can mediate the realization of human subjecthood, and it can also weaken its conditions, but it does not replace humanity as the risk-bearing subject in the sense used here.

2.4 Species Continuity Is Not Equivalent to Continuity of Civilizational Conditions

The continuity of the human species is maintained primarily through biological reproduction, intergenerational continuation, and population stability. As long as humans can still

reproduce, survive, and maintain their basic biological structure, continuity in the species-level sense remains. Continuity of civilizational conditions is different. It depends on the transmission of language, memory, education, institutions, knowledge, norms, technology, structures of responsibility, and narratives of meaning. It requires not only that the next generation be born, but also that the next generation enter a common world that can be understood, inherited, criticized, and renewed.

There is therefore no automatic equivalence between species continuity and continuity of civilizational conditions. A society may continue in the biological sense while undergoing a severe rupture in civilizational conditions. People may still be born, work, consume, and entertain themselves, but the common world they enter may no longer stably maintain the transmission of knowledge, responsibility, judgment, and meaning. If such deterioration is irreversible or difficult to reverse, it is not merely ordinary social change; it may become structural damage to the conditions under which human subjecthood is realized.

This means that some risks may not directly kill humans, yet may gradually deprive humans of the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future. We cannot infer that humanity's future remains open simply because the human species still exists; nor can we infer that a harm lacks existential significance simply because it does not appear as physical destruction.

2.5 Section Summary

This section has offered a two-dimensional working definition of “humanity.” Humanity is first *Homo sapiens* as an embodied biological population; species extinction, permanent incapacitation, and future deprivation remain baseline forms of existential risk to humanity. But humanity is not merely a biological population, nor a pure information structure. Humanity as an embodied species also realizes capacities for knowledge, judgment, responsibility, institutional participation, meaning-creation, and future-directed action under civilizational conditions.

It is therefore necessary, but insufficient, to understand AI-related existential risk to humanity only in terms of species survival. The next section further analyzes the civilizational conditions upon which the realization of human subjecthood depends: how human civilization, as an intergenerational common world, enables humanity as an embodied species to become knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects.

3 | Human Civilization: The Intergenerational Condition for the Realization of Subjecthood

The previous section provided a two-dimensional working definition of humanity. This section defines human civilization—the intergenerational condition for the realization of

human subjecthood.

3.1 Civilization and Human Civilization: Scope of the Paper

This paper does not propose a general theory of all possible civilizations. Unless otherwise specified, “civilization” in this paper refers to “human civilization.” Human civilization is the intergenerational common world formed by humanity, as an embodied species, through language, knowledge, institutions, law, education, structures of responsibility, technological practices, and narratives of meaning. It is constituted by symbolic-institutional-technological structures that can be transmitted, revised, and inherited, enabling individuals to enter a historical and interpretive space larger than their biological lives and to exist as knowers, judges, bearers of responsibility, institutional participants, and creators of meaning.

Human civilization does not mean a closed system composed only of individual humans and excluding nonhuman technological components. Writing, tools, machines, cities, legal texts, the Internet, platform systems, and AI systems can all become components or mediating structures of human civilization. Human civilization is “human” not because it contains no nonhuman elements, but because its risk-bearing subject is humanity as an embodied species and its normative focus is the realization of human subjecthood.

Human civilization is therefore not an independent bearer outside humanity, but a constitutive condition for the realization of human subjecthood. It is not an ornament added to the human species, nor a simple sum of population, states, cultures, or technological systems. It is the common world in which humans are able, across individuals, institutions, and generations, to know, judge, take responsibility, transmit, and create meaning.

3.2 Human Civilization Is Not the Sum of Population

Human civilization cannot be reduced to population size. The simultaneous existence of many human individuals does not automatically constitute civilization. Population provides the biological substrate of civilization, but population by itself does not automatically produce transmissible language structures, systems of knowledge, institutional norms, mechanisms of responsibility, or narratives of meaning. Civilization requires some cross-individual, cross-group, and intergenerational structure that enables knowledge, norms, institutions, responsibility, and meaning to be preserved, transmitted, criticized, and renewed.

A society with a large population may still have difficulty maintaining complex civilizational conditions if it lacks stable language, memory, education, institutions, structures of responsibility, and knowledge transmission. Conversely, some civilizational traditions may continue to influence later generations through language, texts, education, rites, institutional memory, religion, art, and philosophical traditions, even after suffering enormous losses in population, territory, or political power. This shows that there is no automatic equivalence between the continuity of civilizational conditions and population continuity.

3.3 Human Civilization Is Not the State

Human civilization also cannot be identified with the state. The state is a political and legal form of organization, usually characterized by territory, sovereignty, administrative structures, legal systems, public finance, and coercive power. Human civilization is a deeper, broader, and longer-term intergenerational common world. A civilizational tradition may cross multiple states, and a state may carry multiple civilizational traditions. Modern states are also widely embedded in global scientific systems, financial systems, international legal systems, technical standards, digital platforms, and transnational supply chains, none of which belongs wholly within a single state.

Modern scientific systems, university institutions, medical knowledge, engineering standards, the language of the rule of law, market mechanisms, international trade, financial infrastructures, and Internet protocols all cross national boundaries in different ways. States can organize, protect, regulate, or distort these systems, but the state itself is not identical with civilization as a whole. Likewise, no particular civilizational tradition can be simply compressed into any contemporary state.

The state can therefore be understood as a concrete organizational form of civilizational conditions within a particular historical, geographic, legal, and political environment. Human civilization is the more general, more durable, and more complex intergenerational common world.

3.4 Human Civilization Is Not Synonymous with Culture

Human civilization should also not be reduced to culture. Culture is an important component of civilization, especially in the form of language, custom, art, aesthetics, ritual, stories, identity, habits, and ways of life. Culture enables memory, value, and meaning within civilization to enter everyday life in forms that can be experienced, expressed, and transmitted. Without culture, civilization would lose its livable interface.

But civilization includes more than culture. It also includes science, law, institutions, education, the economy, technology, public reason, medicine, engineering, administration, infrastructure, and structures of responsibility. If civilization is understood only as culture, we will underestimate those structures within civilization that sustain truth, responsibility, institutions, and knowledge. A society may possess rich culture while remaining seriously fragile in science, law, institutions, or structures of responsibility. Conversely, a highly technologized and institutionalized society may suffer profound emptiness at the levels of culture and meaning.

Culture can therefore be understood as civilization's layer of meaning-expression and its interface with everyday life, but civilization also includes deeper structures of knowledge, institutions, responsibility, and technology.

3.5 Human Civilization Is Not a Technological System

Human civilization likewise cannot be reduced to a technological system. Technology is an important component of civilization, and it often profoundly changes civilizational forms. From agriculture, metallurgy, navigation, printing, electricity, and computers to artificial intelligence, technology continuously expands human capacities for action and forms of organization. Civilization, however, is not merely a set of tools, nor an efficiency system. Civilization also includes value, norms, responsibility, public reason, meaning, and human understanding of the future.

Modern technological society easily equates civilizational progress with growth in capability, increases in efficiency, expansion of information, and higher degrees of automation. But growth in capability is not automatically equivalent to healthy civilizational conditions. A system may become more efficient while responsibility becomes harder to allocate; may generate more content while truth becomes harder to verify; may automate more processes while cultivating less human judgment; may provide more convenience while finding it harder to answer why humans should continue to learn, create, bear responsibility, and protect the future.

Civilization therefore cannot be defined as the sum of technological capacities. Technology is a powerful system within civilization, but technology itself does not provide final purposes, does not define responsibility, does not guarantee truth, and does not automatically generate meaning. Without deeper civilizational conditions, technological growth may bring expanded capabilities, but not necessarily the maturation of the realization of human subjecthood.

3.6 Human Civilization as a Whole: A Common Ecology Formed by Multiple Civilizational Traditions

Human civilization, as used in this paper, is not identical with any one state, culture, or historical civilizational tradition, nor does it universalize a particular civilizational form as all humanity. Human civilization should be understood as an overall civilizational ecology jointly constituted by multiple civilizational traditions, shared institutional interfaces, knowledge exchange, technological infrastructures, experiences of conflict, and intergenerational structures of transmission. Different civilizational traditions may have different languages, religions, institutions, laws, knowledge traditions, artistic forms, and narratives of meaning, but together they constitute the intergenerational common world formed by humanity as an embodied species in history.

If human civilization is understood as the expansion of one civilizational tradition, then AI-related existential risk to humanity will easily be misread as an internal problem of a particular culture, state, or region. If human civilization is understood as a mechanical sum of multiple civilizational traditions, then the shared structures, cross-dependencies, and common vulnerabilities among those traditions will be underestimated. More precisely, human civilization as a whole contains both difference and common interfaces; it preserves plurality while also possessing certain modern global operating conditions.

Modern global civilizational ecology has created an unprecedented degree of mutual embedding among different civilizational traditions. Modern scientific systems, university institutions, international law, public-health systems, global trade, financial networks, Internet protocols, digital platforms, standardized technological systems, and AI infrastructures no longer belong wholly to any single civilizational tradition, state, or culture. They form shared interfaces that operate across civilizations, enabling different traditions to remain mutually dependent in knowledge, technology, institutions, economics, and governance.

This does not mean that human civilization has become a homogeneous whole. Different civilizational traditions continue to preserve distinct historical memories, religious structures, legal traditions, family ethics, political imaginaries, educational practices, and narratives of meaning. They may also provide different answers to foundational categories such as truth, responsibility, subjecthood, value, and meaning. This paper therefore does not claim that there exists a completely unified and undifferentiated human civilization. Human civilization as used here is a civilizational ecology jointly constituted by differentiated traditions and shared interfaces.

3.7 Human Civilization as an Intergenerational Common World

More precisely, human civilization is an intergenerational common world. Through language, memory, institutions, and education, it allows the past to be transmitted to the future, and allows finite individual lives to enter a historical structure larger than themselves. An individual life is finite. A person does not start from scratch at birth. One enters an already existing language, receives existing institutions and knowledge, and inherits historical memories, scientific achievements, legal structures, ethical norms, and narratives of meaning that one did not personally create. Civilization is precisely this intergenerational structure: it enables people to judge, create, and take responsibility on the basis of what predecessors have built, and it enables the actions of one generation to have meaning for future generations.

This intergenerational character gives civilizational conditions vulnerabilities different from those of species continuity. Species continuity depends primarily on biological reproduction; the continuity of civilizational conditions depends on transmission, understanding, and renewal. If the next generation is merely born biologically, but does not enter a common world capable of cultivating judgment, responsibility, truth, and meaning, then the conditions for the realization of human subjecthood will be weakened. The true continuity of civilizational conditions is not simply “there are people alive,” but “there are people capable of understanding, inheriting, criticizing, and renewing a common world.”

Understanding civilization as an intergenerational common world therefore helps identify risks that do not appear in the form of physical disaster, but may nonetheless alter humanity’s capacities as a future-oriented subject. The continuity of civilizational conditions requires not only data preservation, content generation, and institutional operation, but also that humans remain able to enter, understand, judge, and renew this common world.

3.8 Functional Systems of Civilization and Structures of Human Participation

Human civilization is not an abstract whole; it operates concretely through a set of functional systems of civilization. Economic, legal, educational, scientific, state-governance, cultural, media, medical, administrative, infrastructural, religious, and artistic systems constitute the main pathways through which humans actually enter civilizational conditions. Ordinary people usually do not confront “civilization” as an abstract object. They enter the common world through schools, courts, laboratories, markets, media, government agencies, platform systems, families, religious traditions, artistic practices, and public services.

These functional systems of civilization are not external appendages of civilization, but important support structures for the realization of human subjecthood. Through educational systems, humans learn how to understand, judge, and ask questions. Through scientific systems, they participate in knowledge production and the testing of evidence. Through legal systems, they understand responsibility, rights, and duties. Through economic systems, they participate in labor, exchange, and the organization of resources. Through state and governance systems, they participate in public decision-making and institutional direction. Through cultural and media systems, they express meaning, transmit memory, and form public imagination. Human subjecthood does not occur only inside individuals; it is also realized through structures of participation within these systems.

“Structures of participation” refers to the ways in which humans participate in the operation of the common world as workers, learners, judges, creators, citizens, bearers of responsibility, and future-oriented subjects within the functional systems of civilization. If humans retain only formal confirmation in these systems, while no longer genuinely understanding, judging, taking responsibility, creating, or shaping direction, then the realization of human subjecthood under civilizational conditions is weakened.

3.9 Human Civilization as a Structure of Knowledge, Responsibility, and Meaning

Human civilization, as an intergenerational common world, contains at least three functional structures: a knowledge structure, a responsibility structure, and a meaning structure.

First, human civilization provides a knowledge structure. Humans do not face the world alone and independently reinvent knowledge. Humans know the world through language, education, scientific methods, rules of evidence, systems of literature and documentation, recording institutions, peer review, and public discussion. Civilizational conditions enable individuals to share, criticize, preserve, and revise knowledge. Without some stable knowledge structure, humans have difficulty distinguishing fact from rumor, evidence from narrative, explanation from fabrication, and knowledge from opinion.

Second, human civilization provides a responsibility structure. Through civilizational conditions, humans define actors, commitments, obligations, fault, rights, punishment, compensation, and institutional responsibility. Without responsibility structures, complex society cannot maintain trust, nor can it handle harm, error, conflict, and commitment. Respon-

sibility structures depend on conditions such as attributable action, identifiable subjects, interpretable norms, testable evidence, and traceable procedures.

Finally, human civilization provides a meaning structure. Humans do not merely need to live; they also need to understand why they should learn, work, create, bear responsibility, and protect the future. Civilization organizes meaning through religion, philosophy, art, education, family, tradition, political ideals, and public narratives. Meaning is not an optional psychological decoration, but a structure that makes individual action, responsibility, learning, creation, sacrifice, and commitment to the future intelligible.

Knowledge structures, responsibility structures, and meaning structures together show that human civilization is not a simple sum of population, state, culture, or technological system. It is the common world that enables humans to know, judge, take responsibility, transmit, and create meaning. If these structures are systematically weakened, the conditions for the realization of human subjecthood are damaged.

The stable operation of these structures also depends on deeper foundational categories and their calibration mechanism. The next section asks: which foundational categories must the functional systems of civilization invoke? How are these categories clarified, maintained, and revised? This paper calls this mechanism the philosophical kernel: the category-calibration mechanism internal to civilizational conditions.

4 | The Philosophical Kernel: The Category-Calibration Mechanism Internal to Civilizational Conditions

The previous section showed that civilization carries structures of knowledge, responsibility, and meaning. The stable operation of these structures depends on deeper foundational categories and their calibration mechanism. This section asks: which foundational categories must the functional systems of civilization invoke? How are these categories clarified, maintained, and revised?

4.1 Definition and Scope of the Philosophical Kernel

The philosophical kernel, as used in this paper, is not the set of foundational categories themselves, but the category-calibration mechanism internal to civilizational conditions that maintains the operability of foundational categories. It has the property of a meta-mechanism: “meta” here does not mean that it constitutes a higher entity outside civilizational conditions, nor that it becomes a third risk-bearing subject; rather, it means that what it calibrates is not the local rules internal to one functional system of civilization, but the foundational category interfaces jointly invoked by functional systems such as science, law, education, politics, the economy, and culture. “Philosophy” here does not first mean philosophy as an academic discipline, nor only theoretical texts written by professional philosophers. More precisely, the philosophical kernel refers to the operational mechanism through which foundational categories are made explicit, tested, clarified, criticized,

maintained, and revised in history; foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning are its objects of calibration and conceptual interfaces.

The philosophical kernel is not an independent entity within civilization, not an ordinary sector parallel to science, law, politics, education, the economy, culture, and art, and not a risk-bearing subject. It acts upon a set of underlying category interfaces: science needs categories of truth and evidence; law needs categories of subject, intention, responsibility, and attribution; politics needs categories of legitimacy, publicness, and the justification of power; education needs categories concerning the human, growth, knowledge, and purpose; the economy needs categories of value, preference, exchange, and the justification of distribution; culture and art need categories of meaning, expression, and common experience. If these foundational categories lose stability, the functional systems of civilization may continue to operate, but their operation will gradually drift into states that are unintelligible, unaccountable, and uncalibratable.

In this sense, to call philosophy a “kernel” is not rhetorical elevation, but a judgment about functional level. The kernel does not mean that philosophy is always historically prior to civilization, nor that civilization is deduced from a prewritten philosophical theory. It means that, in their operation, the functional systems of civilization inevitably depend on foundational categories concerning truth, evidence, responsibility, subjecthood, value, and meaning. When these categories cannot be clarified, maintained, and revised, civilization does not merely lose one discipline or one cultural tradition; it loses the underlying mechanism that maintains the stability of its foundational categories.

The ontological position of the philosophical kernel must therefore be clearly delimited: it is the category-calibration mechanism internal to civilizational conditions, not an independent dimension placed alongside civilizational conditions, and not a third risk-bearing subject. The risk-bearing subject is always humanity as an embodied species; civilizational conditions are the intergenerational common world in which human subjecthood is realized; the philosophical kernel is the internal mechanism through which foundational categories are clarified, maintained, and revised.

4.2 Foundational Categories and Modern Global Civilizational Ecology

Foundational categories are not an eternal framework that applies a priori to all possible civilizations, nor are they theoretical vocabularies monopolized by a single civilizational tradition. Different civilizational traditions may organize questions of truth, responsibility, subjecthood, value, and meaning through different languages, religions, rites, institutions, political imaginaries, and knowledge traditions. But insofar as a civilization can distinguish truth from falsity, accumulate knowledge, allocate responsibility, organize action, evaluate value, and interpret meaning, it is already invoking some configuration of foundational categories.

Modern global civilizational ecology gives these categories a new commonality. Scientific knowledge, legal responsibility, technological governance, public decision-making, educational transmission, digital platforms, international institutions, and transnational markets

are deeply intertwined. The knowledge system within one state or civilizational tradition often depends on global research networks, technical standards, digital infrastructures, and international institutional interfaces. The deployment of an AI system in one region may also affect other regions through platforms, supply chains, data ecologies, and structures of public discourse. It is in this context that truth, evidence, responsibility, subjecthood, value, and meaning become unavoidable foundational categories in AI risk analysis.

This does not mean that different civilizational traditions understand these categories in exactly the same way. Truth can be organized through different knowledge traditions; responsibility can be allocated through different legal and ethical systems; subjecthood can be understood through different conceptions of personhood, family, community, and politics; value and meaning can also be interpreted through different religious, philosophical, cultural, and historical experiences. The point of this paper is not to erase this plurality, but to point out that, whatever their concrete answers, the knowledge, institutional, technological, and governance systems of modern global civilizational ecology must invoke these interfaces in some way.

Foundational categories are therefore neither purely a priori nor arbitrarily relative. They are functionally necessary: without truth and evidence, knowledge cannot stably accumulate; without responsibility, action cannot be attributed; without subjecthood, judgment, consent, authorization, and responsibility-bearing lose their basis; without value, technical optimization loses its target; without meaning, learning, creation, responsibility, and commitment to the future lose their justificatory structure.

4.3 The Functional Position of Six Foundational Categories

This paper focuses on six foundational categories: truth, evidence, responsibility, subjecthood, value, and meaning. They do not exhaust all foundational concepts of human civilization, but they are sufficient to mark the category-calibration mechanism that concerns this paper.

The category of truth enables civilization to distinguish truth from falsehood, fact from narrative, knowledge from opinion, and explanation from fabrication. Without the category of truth, knowledge production becomes content production, public discussion loses a shared orientation, and institutional judgment lacks testable objects.

The category of evidence answers what can support a judgment, what can refute a judgment, and what can enter the public exchange of reasons. Without the category of evidence, science, law, education, media, and public governance have difficulty stably distinguishing the reliable from the unreliable, the verifiable from the unverifiable, and inference from guesswork.

The category of responsibility connects action, subject, authorization, intention, fault, consequence, and obligation. Without the category of responsibility, society may still produce outcomes and consequences, but it becomes increasingly difficult to ask who acted, who authorized, who knew, who could have prevented, and who should bear the consequences.

The category of subjecthood answers who can judge, consent, refuse, authorize, act, bear

responsibility, and represent themselves. Without the category of subjecthood, law, politics, education, ethics, and institutional participation lose their basis, because all these systems require some understanding of the capacities and qualifications of subjects.

The category of value enables civilization to rank safety, freedom, efficiency, truth, dignity, justice, innovation, stability, prosperity, and meaning. Without the category of value, technical systems can continue to optimize, but civilization will have difficulty explaining what optimization is supposed to serve.

The category of meaning answers why we learn, why we create, why we bear responsibility, why we protect the future, and why humans still matter. Without the category of meaning, society can continue to produce, consume, entertain, and optimize, but human action, responsibility, and commitment to the future become increasingly difficult to explain in common terms.

Together, these six foundational categories constitute the underlying interfaces that the functional systems of civilization must invoke. They are not isolated words, but a mutually supporting category network. Truth requires evidence for calibration; responsibility requires subjecthood as its bearer; value requires meaning for interpretation; and meaning requires truth, responsibility, and value if it is not to collapse into arbitrary narrative. The function of the philosophical kernel is precisely to maintain the distinguishability, connectability, and revisability of these categories.

4.4 How the Category-Calibration Mechanism Works

The category-calibration mechanism refers to the process through which foundational categories are made explicit, tested, criticized, maintained, and revised in history. It does not mean that there is a single center capable of uniformly managing all categories, nor that one discipline can monopolize the authority to interpret them. It means that, under civilizational conditions, foundational categories can be continually clarified and revised through philosophy, science, law, public discussion, education, institutional reform, cultural practice, and historical experience, thereby preventing functional systems from operating for long periods on miscalibrated categories.

Category calibration differs from local error correction. Local error correction can address a model error, a legal loophole, a policy failure, or an educational problem. Category calibration addresses more foundational questions: what counts as an error, what counts as evidence, who can bear responsibility, what constitutes judgment, what is worth optimizing, and what remains meaningful. If these questions cannot be posed and addressed, the functional systems of civilization may continue to operate, but increasingly by inertia, efficiency, and automation rather than by understanding, judgment, and responsibility.

Category calibration also differs from mere value declaration. A society can repeatedly declare that it values truth, responsibility, dignity, or meaning, but if these categories cannot be institutionalized, educated, made public, and tested and revised in practice, such declarations cannot be transformed into stable civilizational conditions. Category calibration requires that foundational categories be continually invoked, questioned, and updated

within concrete social mechanisms.

The philosophical kernel is therefore not a storehouse of abstract ideas, but a dynamic mechanism within civilizational conditions that maintains the operability of foundational categories. It exists in theoretical reflection, but also in legal procedure, scientific norms, educational training, public debate, institutional reform, and cultural practice. Its function is not to keep civilization fixed at one set of answers, but to enable civilization, when facing new technologies, new institutions, and new risks, to clarify anew the basic concepts it invokes.

If the calibration mechanism for foundational categories fails, the functional systems of civilization may continue to operate, but they may gradually lose the basis of their intelligibility, accountability, and capacity for correction.

4.5 The Mapping Relation among Foundational Categories, Foundational Civilizational Conditions, and Capacities of Subjecthood

To avoid conceptual drift, it is necessary to distinguish three related but non-identical lists: foundational categories, foundational civilizational conditions, and capacities of subjecthood.

First, foundational categories are the conceptual interfaces that the functional systems of civilization must invoke. This paper focuses on truth, evidence, responsibility, subjecthood, value, and meaning. They answer how a civilizational system conceptually distinguishes and organizes the world.

Second, foundational civilizational conditions are the conditions under which these foundational categories remain operable within the functional systems of civilization. This paper focuses on five types: the calibrability of truth and evidence; the attributability of responsibility and authorization; the sustainability of subjecthood and judgment; the openness of value-ordering to reflection; and the intelligibility and continuity of meaning-transmission. They answer whether these foundational categories can still be stably invoked, tested, and revised in civilizational practice.

Third, capacities of subjecthood are the capacities actually exercised by humans under civilizational conditions. This paper focuses on six types: knowing what is true, forming judgment, bearing responsibility, participating in and sustaining institutions, transmitting and creating meaning, and acting toward the future. They answer whether humans can still exist as knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects.

These three lists are not independent parallel lists. More precisely, foundational categories provide conceptual interfaces for foundational civilizational conditions; foundational civilizational conditions support the actual exercise of capacities of subjecthood. If truth and evidence cannot be calibrated, humans have difficulty knowing what is true and forming judgment. If responsibility and authorization cannot be attributed, humans have difficulty bearing responsibility and sustaining institutions. If subjecthood and judgment cannot be sustained, humans have difficulty participating in the common world as actors and future-oriented subjects. If value-ordering is not open to reflection, technological and institutional

optimization may become detached from human purposes. If meaning-transmission cannot be understood and continued, humans lose the justificatory structure for learning, creating, taking responsibility, and protecting the future.

This mapping relation also shows that the paper is not simply listing abstract concepts, but explaining the conditional structure of the realization of human subjecthood. Foundational categories are conceptual interfaces; foundational civilizational conditions are operating conditions; capacities of subjecthood are the actual realization of humans within civilization.

4.6 Disambiguating the “Category of Subjecthood” and the “Realization of Subjecthood”

This paper uses the term “subjecthood” repeatedly, so it is necessary to clarify its two distinct levels of meaning.

As a foundational category, subjecthood refers to the conceptual structure concerning who can judge, consent, authorize, act, take responsibility, and represent themselves. It belongs to the category-calibration mechanism. Legal systems, political systems, educational systems, ethical relations, and public governance all depend on some category of subjecthood in order to determine who can sign, who can be responsible, who can be authorized, who can consent or refuse, and who has the capacity for action and responsibility.

As part of the realization of human subjecthood, subjecthood refers to the actual human exercise of capacities such as knowing what is true, forming judgment, bearing responsibility, participating in and sustaining institutions, transmitting and creating meaning, and acting toward the future. It is not an abstract concept, but the practical capacity through which humans unfold their mode of existence under civilizational conditions. Humanity is not merely a biological population precisely because humans are capable, under civilizational conditions, of knowing what is true, forming judgment, bearing responsibility, participating in and sustaining institutions, transmitting and creating meaning, and acting toward the future.

These two levels must not be confused. Subjecthood as a foundational category is a conceptual condition for the realization of human subjecthood. The realization of subjecthood as an actual capacity is the practical unfolding of those conceptual conditions within human civilization. The former is not a repetition of the latter; it is the conceptual precondition through which the latter can be understood, institutionalized, and maintained.

This disambiguation is important for AI risk analysis. AI may weaken the actual capacities through which human subjecthood is realized, and it may also make the category of subjecthood itself obscure. The former concerns the weakening of capacities of subjecthood; the latter concerns the miscalibration of the category of subjecthood. The two are connected, but they must not be conflated.

4.7 Section Summary

This section has defined the philosophical kernel as the category-calibration mechanism internal to civilizational conditions. The philosophical kernel is not an independent entity within civilization, not a third layer parallel to human civilization, and not a risk-bearing subject. The risk-bearing subject is always humanity as an embodied species. The function of the philosophical kernel is to explain how human civilization maintains the intelligibility, accountability, and revisability of foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning.

This section has also distinguished three lists: foundational categories, foundational civilizational conditions, and capacities of subjecthood. Foundational categories provide conceptual interfaces for foundational civilizational conditions; foundational civilizational conditions support the actual exercise of capacities of subjecthood. Through this mapping relation, the paper avoids reifying the philosophical kernel and avoids treating civilizational conditions as an abstract background.

The next section uses a functional analogy to translate the relations among the human species, human civilization, the philosophical kernel, and the functional systems of civilization into a more intuitive structural diagram. This analogy is not an ontological model, but a tool for helping readers understand the relations among biological substrate, category interfaces, civilizational operating environment, and application-layer functional systems.

5 | Hardware—Kernel—Distribution—Application Layer: A Functional Analogy for Understanding the Structural Relations

The previous sections have explained the positions of the human species, human civilization, the philosophical kernel, and the functional systems of civilization. These relations are relatively abstract, so this paper introduces a functional analogy to help explain them.

5.1 Limited Use and Boundaries of the Analogy

It must first be made clear: humans are not machines, philosophy is not code, and civilization is not software. The hardware—kernel—distribution—application-layer analogy is not used in order to technologize human civilization, nor to replace philosophical analysis with a computer-system model. It is only a limited pedagogical metaphor for explaining four functional relations. [Note 5]

First, human civilization needs an embodied biological substrate, just as a computing system needs physical hardware. Second, the functional systems of civilization need foundational category interfaces, just as applications need underlying system services. Third, humans actually live in a complete civilizational operating environment, not directly in abstract categories; this is similar to the way ordinary users typically enter a complete distribution rather than directly facing the kernel. Fourth, the functional systems of civilization—

science, law, education, the economy, politics, culture, media, administration, and so on—can be roughly compared to the application layer, because they are the main interfaces through which humans actually enter the common world in everyday life.

This analogy helps avoid two misunderstandings. The first is the assumption that the survival of the human species is equivalent to the completeness of the conditions for the realization of human subjecthood. The second is the assumption that the continued operation of the functional systems of civilization means that the underlying categories remain healthy. In fact, a system may continue to run on the surface, and may even run faster, more efficiently, and more automatically, while its underlying interfaces have become unintelligible, unaccountable, or uncalibratable.

All analogies in this section must therefore obey one restriction: they explain only functional relations, not ontological relations. Non-analogically speaking, the risk-bearing subject is always humanity as an embodied species. Civilization, the philosophical kernel, functional systems, and AI are not risk-bearing subjects parallel to humanity.

5.2 The Human Species as Hardware: Biological Substrate

Within the structural relations of human existence, the human species provides the most basic biological substrate. In the computer-system analogy, it is similar to hardware. Hardware provides the material basis for system operation. Without processors, memory, storage, power supply, and input-output devices, no kernel, distribution, or application can actually run. Similarly, without human bodies, brains, nervous systems, capacities for perception, reproductive mechanisms, mortality, and ecological dependence, human civilization cannot continue to unfold.

This analogy helps explain the baseline status of species-level risk. Human civilization is not an information cloud floating free from bodies. Language learning, educational transmission, emotional development, ethical relations, responsibility-bearing, and meaning-formation all depend on embodied human life. If the human species becomes extinct, or if humanity permanently loses the capacity for continuation and action, then civilizational conditions, the category-calibration mechanism, and functional systems will lose their bearer.

But hardware is not a complete operating environment. A computer that has only hardware but no kernel, file system, user environment, or applications is not a usable system. Similarly, the existence of humanity as a biological species does not automatically produce science, law, education, institutions, structures of responsibility, historical memory, or narratives of meaning. Species survival is a necessary condition, but not a sufficient condition.

The analogy of the human species as “hardware” therefore explains only the function of biological substrate. It must not reduce humans to machine components. Humanity as an embodied species has mortality, vulnerability, historicity, and relationality, none of which can be adequately expressed by the concept of computer hardware.

5.3 The Philosophical Kernel as Kernel: Calibration of Category Interfaces

If the human species can be analogized to hardware, then the philosophical kernel can be analogized to the kernel. A kernel is not the complete system that users operate directly in everyday life, but it provides underlying interfaces for upper-level systems: resource scheduling, permission boundaries, system calls, process management, and hardware abstraction. Applications usually do not directly face hardware; they run through the interfaces provided by the kernel. If the interfaces at the kernel level become seriously miscalibrated, the application layer may still launch, but system stability, controllability, and reliability are affected.

In human civilization, the philosophical kernel does not mean that a fixed philosophical theory governs civilization, nor that academic philosophy is institutionally superior to science, law, politics, education, the economy, or culture. It refers to the calibration mechanism internal to civilizational conditions that maintains the operability of foundational categories, rather than the set of foundational categories themselves. Science needs interfaces of truth and evidence; law needs interfaces of subject, responsibility, and attribution; politics needs interfaces of legitimacy and publicness; education needs interfaces of knowledge, growth, and purpose; the economy needs interfaces of value, preference, and the justification of distribution; culture and art need interfaces of meaning, expression, and common experience.

Analogizing the philosophical kernel to the kernel helps explain its function as an underlying interface. It does not produce all civilizational content, but it affects whether the functional systems of civilization can stably invoke foundational categories. If the interfaces of truth and evidence are miscalibrated, science and education may continue to produce content, but the connection between knowledge and the world becomes fragile. If the interfaces of responsibility and authorization are miscalibrated, law and administration may continue to run procedures, but the attribution between actions and consequences becomes unstable.

This analogy, however, must not be over-reified. The philosophical kernel is not code, nor a module that can be separately installed, replaced, or restarted. Foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning exist in language, institutions, education, law, science, public debate, and historical experience. The philosophical kernel is called a “kernel” only because it functionally performs the calibration of foundational category interfaces, not because it has the physical or programmatic structure of a computer kernel.

5.4 Human Civilization as a Distribution: The Complete Operating Environment

In a computer system, a distribution organizes system tools, libraries, configurations, permission structures, user interfaces, and maintenance ecosystems around the interfaces provided by the kernel, thereby forming a usable operating environment. This analogy does not require us to understand the distribution-kernel relation as a strict ontological containment relation. It only explains that upper-level operating environments need to invoke underlying interfaces, and that underlying interfaces become usable for users only through

concrete operating environments and application-layer practices.

In this paper, human civilization can be analogized to a complete operating environment. Human civilization is not foundational categories themselves, nor any single institution or technological system, but the intergenerational common world jointly constituted by language, knowledge, law, education, the economy, politics, culture, technology, structures of responsibility, historical memory, and narratives of meaning. The philosophical kernel, as the category-calibration mechanism internal to civilizational conditions, maintains the foundational category interfaces invoked by the functional systems of civilization. This analogy explains only relations of invocation, dependence, and interface; it does not prescribe ontological containment. Individual humans enter civilization not by directly entering abstract categories, but by entering concrete environments constituted by families, schools, law, media, scientific communities, work organizations, cultural traditions, technological platforms, and public institutions.

This analogy helps explain the relation between human civilization and the philosophical kernel. Human civilization is not a container for the philosophical kernel, but a complete operating environment that invokes foundational category interfaces. The philosophical kernel calibrates the foundational category interfaces invoked by the functional systems of civilization; human civilization is the historical operating environment formed by these interfaces, institutions, practices, technologies, memories, and narratives of meaning. Humans do not live in abstract categories, but in this complete operating environment, where they learn, judge, take responsibility, create meaning, and act toward the future.

The distribution analogy also shows that an operating environment may appear complete while its underlying interfaces are miscalibrated. A distribution may still boot, its interface may still respond, and applications may still run, but if its underlying permissions, resource scheduling, or system calls develop serious problems, the system gradually becomes unreliable. Similarly, a society may still have schools, courts, media, markets, government, and cultural production, and these systems may even run more efficiently; but if underlying interfaces such as truth, evidence, responsibility, subjecthood, value, and meaning become miscalibrated, the conditions for the realization of human subjecthood may still be structurally weakened.

The analogy of human civilization as a distribution is therefore not meant to turn civilization into software. It is meant to show that humans actually live in an operating environment composed of multiple systems, institutions, practices, and structures of meaning. This environment depends on embodied humans, on foundational category interfaces, and on the operation of concrete functional systems of civilization.

5.5 Functional Systems of Civilization as the Application Layer

In this analogy, systems such as science, law, politics, education, the economy, culture, media, administration, medicine, religion, and art can be roughly compared to the application layer. The application layer handles concrete tasks directly: producing knowledge, adjudicating disputes, organizing power, educating the next generation, allocating resources, expressing meaning, treating illness, providing services, disseminating information, main-

taining ritual, and interpreting life. What humans most often encounter in everyday life is not foundational categories themselves, nor “civilization” as an abstract whole, but these functional systems.

Ordinary people usually do not enter civilization by reading philosophical definitions. They enter it by learning knowledge in schools, learning relations and responsibilities in families, understanding rights and duties through courts, participating in labor and exchange through markets, understanding evidence and method through scientific systems, encountering public narratives through media, participating in public affairs through state and governance systems, and understanding meaning through culture, religion, and art. Functional systems of civilization are the application-layer interfaces for the realization of human subjecthood.

The application layer is visible, evaluable, and optimizable. Whether a system operates, whether tasks are completed, whether processes accelerate, and whether users are satisfied are usually relatively easy to observe. Modern AI also most readily displays its value at the application layer: raising productivity, reducing cost, assisting writing, automating analysis, accelerating workflows, personalizing services, and lowering barriers to knowledge.

Application-layer prosperity, however, is not equivalent to the health of the system as a whole. A society may produce more reports, more images, more analyses, more recommendations, more evaluations, and more decisions, while at the same time finding it increasingly difficult to confirm the relation between these outputs and truth, responsibility, judgment, and meaning. A functional system of civilization may also continue to operate, and may even operate more efficiently, while gradually requiring less human understanding, judgment, labor, creation, public participation, and responsibility-bearing. In that case, the conditions for the realization of human subjecthood within it are weakened.

The analogy of functional systems of civilization as the application layer therefore provides an intermediate bridge for the AI risk analysis that follows. Humans are not merely users, consumers, or recipients of services in the functional systems of civilization. They should also be workers, learners, judges, creators, citizens, bearers of responsibility, and future-oriented subjects. The next section further explains how AI may change the structures of human participation at the application layer, or destabilize humans’ grasp of the calibration of foundational categories.

5.6 Explaining the Structural Relations Non-Analogically

Non-analogically speaking, the human species, human civilization, the philosophical kernel, and the functional systems of civilization are not four parallel entities and not four risk-bearing subjects. They explain different conditions through which the same risk-bearing subject—humanity as an embodied species—exists and realizes subjecthood.

The human species provides the biological substrate. Without embodied human life, there is no human civilization, realization of subjecthood, or commitment to the future. Human civilization provides the intergenerational common world for the realization of subjecthood. Without language, knowledge, institutions, structures of responsibility, histor-

ical memory, and narratives of meaning, humans have difficulty existing as knowers, judges, bearers of responsibility, and creators of meaning. The philosophical kernel, as the category-calibration mechanism internal to civilizational conditions, maintains the intelligibility, accountability, and revisability of foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. Functional systems of civilization are the application layer through which humans actually participate in these conditions and enter the common world through science, law, education, the economy, culture, governance, and public services.

AI-related existential risk to humanity therefore cannot be understood only as the question of whether some technological system harms human bodies. Nor can it be understood only as whether some social application produces a local problem. AI may act upon the biological substrate, alter the civilizational operating environment, weaken structures of human participation in functional systems, or disturb the calibration mechanism of foundational categories. All these risks must ultimately return to one question: can humanity as an embodied species still realize its subjecthood under civilizational conditions?

The ultimate purpose of the analogy in this section is to help readers understand this structural relation. Hardware—kernel—distribution—application layer is not a new ontological framework, but an explanatory tool. It shows that the realization of human subjecthood requires bodily substrate, underlying category interfaces, a complete civilizational operating environment, and concrete functional systems. If any layer is systematically weakened, humanity's capacity as a future-oriented subject may be affected; but the risk-bearing subject is never these layers themselves. It is humanity as an embodied species.

The next section turns to the paper's central definitional question: why does AI make these conditions matters of existential risk? How may AI act upon biological survival, civilizational conditions, and the category-calibration mechanism? Under what thresholds do such actions constitute AI-related existential risk to humanity?

6 | Why AI Makes These Conditions Matters of Existential Risk: Expanded Definition and Thresholds

The preceding sections have explained that the risk-bearing subject of AI-related existential risk to humanity is always humanity as an embodied species. The possible modes of harm to humanity include not only destruction or permanent incapacitation at the biological-survival layer, but also the irreversible or difficult-to-reverse weakening of the foundations for the realization of subjecthood at the civilizational-conditions layer. Within the civilizational-conditions layer, AI may also disturb the calibration mechanism for foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning. On this basis, this section answers: how may the development, deployment, and deep integration of AI act upon these layers and mechanisms? Under what thresholds do such effects become not merely ordinary social risks, but enter the scope of AI-related existential risk to humanity?

6.1 Boundary of This Section: Not a Complete Risk Classification

This section is not a complete classification of AI risk types, nor an elaboration of subsequent governance proposals. Its task is narrower: to explain why AI may touch the foundational conditions of existential risk to humanity, and to provide the expanded definition and thresholds of judgment.

This section does not treat “AI and biological survival,” “AI and civilizational conditions,” and “AI and category calibration” as three independent risk types. They are ways of explaining how the same risk-bearing subject—humanity as an embodied species—may be harmed at different layers or mechanisms. Likewise, this section does not call every social change caused by AI an existential risk. [Note 6] Only when a change acts upon the foundational conditions for the realization of human subjecthood and has cross-domain, structural, and irreversible or difficult-to-reverse consequences does it enter the expanded scope of existential risk in this paper.

6.2 AI and the Biological-Survival Layer

First, AI may act upon the biological-survival layer of humanity. This is the most familiar part of existing discussions of AI existential risk, and it remains the baseline form of risk preserved throughout this paper. If AI systems, through loss of control, misuse, competitive pressures, weaponization, high-consequence technological chains, or critical-infrastructure risks, lead to human species extinction, permanent incapacitation, or irreversible closure of humanity’s future, then they directly constitute existential risk to humanity.

Such risk may arise through multiple pathways. Sufficiently powerful AI systems may assist dangerous biotechnology research, enhance cyberattack capabilities, affect critical infrastructures such as energy, communication, finance, transportation, and supply chains, participate in automated weapons systems, or amplify cascading consequences in complex social systems that humans cannot understand or control in time. This paper does not elaborate these specific pathways here, because it is not a risk taxonomy. But these possibilities are sufficient to show that species-level risk remains a core component of AI-related existential risk to humanity.

The importance of the biological-survival layer also lies in its status as the bearer of all other layers and mechanisms. Without embodied human life—without bodies, nervous systems, mortality, reproductive mechanisms, and intergenerational continuation—human civilization, the philosophical kernel, responsibility structures, and narratives of meaning cannot continue. The paper’s emphasis on civilizational conditions and category calibration therefore does not weaken species-level risk. On the contrary, it takes species-level risk as the baseline and further explains that humanity’s future depends not only on whether bodies survive, but also on whether humans can still exist as future-oriented subjects under civilizational conditions.

Moreover, civilizational conditions and the category-calibration mechanism themselves affect humanity’s ability to respond to species-level risk. A society that cannot stably dis-

tinguish truth from falsehood or evidence from narrative has difficulty reliably assessing high-consequence technological risks. A society with unclear responsibility structures has difficulty governing dangerous chains of action. A society whose capacities for subject judgment have weakened has difficulty forming long-term public action. A society with impoverished meaning structures may lack reasons to protect future generations. Species-level risk and risk to the conditions for the realization of subjecthood are therefore not competitors. They are mutually dependent. The former provides the baseline; the latter explains how humanity's future above that baseline can still be a human future.

6.3 AI and the Civilizational-Conditions Layer: Deprivation by Substitution

AI's impact on the civilizational-conditions layer is not merely that some tool becomes more efficient, nor merely that some industry becomes automated. The deeper issue is that AI is entering the functional systems of civilization and changing how humans participate in the common world.

As Sections 3 and 5 showed, economic, legal, educational, scientific, state-governance, cultural, media, administrative, and public-service systems are important support layers for the realization of human subjecthood. Through these systems, humans learn, work, judge, create, take responsibility, govern, and act toward the future. AI, as a more efficient, more replicable, and more scalable substitute, may gradually replace humans in roles of labor, cognition, creation, governance, and judgment within these systems, pushing humans to the margins of the system or displacing them from it. This paper calls this mechanism **deprivation by substitution**. [Note 7]

Economic systems may depend less and less on human labor and professional judgment; cultural systems may depend less and less on human creation and interpretation; governance systems may depend less and less on human public judgment and political participation; educational and research systems may increasingly depend on automated generation, automated evaluation, and automated filtering. At that point, the functional systems of civilization continue to operate, and may even operate more efficiently, but humans' position as participants in, and shapers of the direction of, those systems is weakened.

This risk does not necessarily appear as the immediate loss of all rights or resources for humans. A more common form may be **relative disempowerment**: humans still consume, vote, confirm, and receive services, but increasingly have less substantive influence over resource allocation, institutional direction, knowledge production, and cultural narratives. Humans remain in the system, but they are no longer necessary participants in its operation. If this trend crosses multiple functional systems—such as the economy, culture, state governance, education, and media—and is reinforced through feedback among them, it may weaken humanity's capacity as a future-oriented subject to shape the common world.

The core question of deprivation by substitution is therefore: can humans still substantively participate in the formation and renewal of the common world as knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects? If the answer gradually becomes negative, then AI's impact on civilizational conditions is not merely a problem of social adaptation. It may touch the foundational conditions

for the realization of human subjecthood.

6.4 AI and Mediated Reconfiguration within Civilizational Conditions

AI's impact on civilizational conditions is not only that humans are gradually replaced or pushed out of the functional systems of civilization. A more subtle mechanism is that humans remain within the system and still retain forms of choice, approval, signature, learning, creation, and participation, while the preconditions of these activities have become deeply mediated by AI. This paper calls this mechanism **mediated reconfiguration**.

Mediated reconfiguration first occurs within the functional systems of civilization. Educational systems still educate students, but learning pathways, question generation, answer evaluation, and knowledge organization are increasingly mediated by AI. Legal and administrative systems still retain procedures, but evidence screening, text generation, risk ranking, and decision recommendations increasingly involve AI. Scientific and knowledge-production systems still publish research, generate explanations, and organize literature, but hypothesis generation, material screening, argumentative writing, and credibility assessment increasingly depend on model outputs. Cultural and media systems still produce meaning, narratives, and public imagination, but forms of expression, allocation of attention, and evaluation of meaning are increasingly shaped by generative systems.

In these situations, humans are not completely excluded from the system. Humans still read, confirm, vote, sign, learn, create, manage, and bear responsibility. The problem is that the structure of human participation has changed. Humans still receive information, but points of entry, summary structures, and interpretive frames may be pre-organized by systems. Humans still make decisions, but visible options, risk rankings, and generated reasons may be shaped by AI. Humans still bear responsibility, but the process by which judgment is formed becomes increasingly difficult to understand and trace. Humans still learn and create, but learning processes, standards of creation, and evaluations of meaning increasingly depend on system output. [Note 8] In this case, subjecthood is not directly abolished, but reorganized beneath the surface of continued participation. [Note 9]

The deeper consequence of mediated reconfiguration is that the way the functional systems of civilization invoke foundational categories changes. It does not merely change how some task is completed. It may also change how foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning are understood, invoked, and calibrated in practice. More precisely, mediated reconfiguration can be understood through five types of pressure on category interfaces and their overall linkage.

First, pressure on the interfaces of truth and evidence. Generative systems can produce fluent, coherent, and apparently credible content, but fluency is not truth, explanation is not evidence, and relevance is not reliability. If humans increasingly take generated content as the starting point of knowledge rather than taking sources, evidence, and verification processes as the basis of knowledge, then the calibration relation between truth and evidence becomes fragile. The problem is not merely that some model occasionally errs, but that the conditions under which human communities maintain verifiable knowledge structures may be altered. [Note 10]

Second, pressure on the interfaces of responsibility and authorization. When AI participates in generating recommendations, screening evidence, assessing risk, comparing plans, and triggering action, the fact that a human signs the final document does not necessarily mean that the human understands the process by which the judgment was formed. The fact that an institution retains approval and audit procedures does not necessarily mean that it can trace how action came into being. Responsibility structures therefore cannot rely only on formal signatures, procedural nodes, or logs. They must clarify anew the relations among action, authorization, understanding, fault, and attribution. Here “authorization” is not an additional foundational category, but a key interface of the category of responsibility in institutional practice.

Third, pressure on the interfaces of subjecthood and judgment. Subjecthood cannot be simply equated with a final click, confirmation, or choice. If AI systems pre-organize problems, filter information, generate reasons, rank options, and recommend actions, then humans may retain the form of choice while participating less and less in the core process through which judgment is formed. Here “judgment” is not an additional foundational category, but a core capacity through which subjecthood is realized in practice. The problem is not whether humans still use tools, but whether tools begin to replace the processes through which humans form reasons, understand problems, and bear judgment. [Note 11]

Fourth, pressure on the interface of value-ordering. AI often operates in the form of optimization problems, but optimization itself does not explain whether the goal is justified, nor whether the optimized metric truly represents human value. A system can optimize engagement while weakening public reason; optimize productivity while damaging subjecthood; optimize convenience while reducing judgment; optimize satisfaction while concealing the poverty of meaning. If the functional systems of civilization increasingly rely on measurable metrics to organize action, value-ordering may be replaced by technical indicators, while value reflection is compressed by optimization logic.

Fifth, pressure on the interface of meaning-generation. AI can generate works, companionship, research, designs, and plans, causing humans to ask again about the foundations of creation, learning, labor, dignity, contribution, and meaning. If civilizational conditions cannot re-explain why humans should still learn, create, bear responsibility, and protect the future, then the realization of subjecthood loses an important support in meaning. The risk here is not whether AI can generate valuable content, but whether humans can still understand the meaning of their own creation, responsibility, and commitment to the future under conditions in which generated content is abundant.

These pressures on interfaces are not isolated from one another. Truth requires evidence for calibration; responsibility requires subjecthood as its bearer; subjecthood requires judgment for its realization; value requires meaning for interpretation; and meaning requires truth, responsibility, and value if it is not to collapse into arbitrary narrative. When AI simultaneously changes the practical conditions of multiple category interfaces, the functional systems of civilization may continue to operate, and may even operate more efficiently, but their operation will increasingly depend on automation, prediction, ranking, and optimization rather than understanding, judgment, responsibility, and meaning. This paper calls the consequence of such linked multi-interface change **systemic drift of the category network**.

Deprivation by substitution and mediated reconfiguration are not mutually exclusive. Humans may be gradually replaced in some functional systems of civilization while still participating in other systems under deep mediation. The former weakens the position of humans as participants in the common world; the latter weakens human understanding, judgment, responsibility, and meaning-creation within the process of participation. Both point to the same question: can humans still realize their subjecthood under civilizational conditions?

6.5 Expanded Definition and Its Necessity

On the basis of the preceding analysis, this paper proposes the following expanded definition:

AI-related existential risk to humanity refers to the risk, arising from the development, deployment, integration, or loss of control of AI systems, that AI may irreversibly destroy the human species; or render humanity permanently incapacitated, or otherwise permanently foreclose humanity's future; or undermine—irreversibly or in ways that are difficult to reverse—the foundational conditions under which humanity, as an embodied species, realizes its subjecthood within civilization.

These three disjuncts are not mutually exclusive categories, but a structure of coverage organized by the question “what continues to exist?” The first refers to the biological termination of the risk-bearing subject itself. The second refers to a case in which the risk-bearing subject continues to exist, but its future is permanently closed; permanent incapacitation is one of its core forms. The third refers to a case in which the risk-bearing subject and a nominal future continue to exist, but the foundational conditions under which that subject realizes its subjecthood within civilization are weakened irreversibly or in ways that are difficult to reverse. The disjuncts may entail and overlap with one another, because the definition aims to jointly cover the possible modes of harm to humanity rather than to produce a formally mutually exclusive classification.

This definition preserves the central status of species-level existential risk, while extending risk analysis from the single biological-survival layer to the realization of subjecthood under civilizational conditions. It does not require that all social changes caused by AI be included within existential risk, nor that every civilizational disturbance be regarded as existential harm. It claims only that when AI's effects touch the foundational conditions under which humanity's realization of subjecthood within civilization can continue to hold, and when those effects have irreversible or difficult-to-reverse structural consequences, theories of existential risk cannot take biological survival as their only object of judgment.

This risk need not appear as sudden collapse. It may appear as a transfer of capacities within normal operation: humans still receive information, but verify sources less and less; still make choices, but participate less and less in defining the problem; still bear responsibility, but understand less and less the process by which judgment is formed; still produce content, but find it increasingly difficult to maintain the relation between truth and evidence; still possess institutions, but find it increasingly difficult to trace chains of responsibility between action and consequence. Society may then become more efficient, more convenient, and more automated, while the conditions for the realization of human

subjecthood are gradually weakened.

If theories of existential risk focus only on biological destruction, they will therefore miss a risk form distinctive to the AI era: the human species continues to exist, but the civilizational conditions under which humans continue to exist as knowers, judges, bearers of responsibility, institutional participants, and creators of meaning are systematically weakened. The expanded definition proposed here is intended to provide a conceptual place for this risk.

6.6 Four Thresholds for Risk to the Realization of Human Subjecthood

For a risk to enter the scope of existential risk at the level of the realization of human subjecthood, it should satisfy at least four conditions. Serious social problems that do not satisfy these thresholds may still be important risks, but they are not necessarily existential risks. A qualification is needed at the outset: the following four thresholds primarily apply to risks to the realization of human subjecthood at the civilizational-conditions layer. Baseline risks at the biological-survival layer, such as species extinction, permanent incapacitation, or permanent foreclosure of humanity's future, directly constitute existential risks by their own nature and do not need, nor are they suited to, assessment through these four thresholds. The function of the four thresholds is to judge AI risks that do not directly appear as species destruction but may weaken—irreversibly or in ways that are difficult to reverse—the conditions for the realization of human subjecthood.

First, it must act upon foundational civilizational conditions rather than local functions. The replacement of a profession, the restructuring of an industry, or the automation of an institutional process does not automatically constitute an existential risk. Only when these changes systematically affect human capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future can they enter the scope of existential risk at the level of the realization of human subjecthood.

Second, it must possess cross-domain diffusion capacity. A local error, single accident, or isolated misuse is usually insufficient to constitute existential risk. Only when a mechanism can cross education, science, law, media, business, public governance, cultural production, and everyday life, repeatedly changing structures of human judgment, responsibility, participation, or meaning across multiple systems, does it have the risk-character of affecting the realization of human subjecthood.

Third, it must be capable of causing irreversible or difficult-to-reverse structural damage. Existential risk concerns not only scale, but also reversibility. Here “irreversible or difficult to reverse” must be understood in layers. At the species level, it primarily refers to absolute biological or physical irreversibility, such as human species extinction, rupture of the reproductive chain, or permanent incapacitation. At the civilizational-conditions layer, it primarily refers to difficulty of reversal in historical, institutional, and cognitive senses. For example, long-term institutional path dependence may turn certain automated processes into default structures; breaks in intergenerational capacity formation may leave later generations without certain capacities for judgment, verification, and responsibility-

bearing; deep reshaping of cognitive habits may lead people no longer to recognize the absence of certain capacities as a loss. This kind of irreversibility differs from the physical nonrecoverability of species extinction, but over historical time it may be sufficient to change humanity's capacity to continue as a future-oriented subject. [Note 12]

Fourth, it must cause the harms identified in the first three thresholds to converge into overall damage to humanity's capacity as a future-oriented subject. Existential risk is not any major loss, but a risk that affects whether humanity can continue as the subject of its own future. The fourth threshold does not repeat the first. Rather, it tests whether the first three conditions jointly converge into overall, intergenerational damage to future-oriented capacity. If a mode of AI integration not only acts upon foundational civilizational conditions, diffuses across domains, and causes irreversible or difficult-to-reverse structural damage, but also makes it increasingly difficult for humans to understand risks, revise institutions, organize action, and shape the future, then it is not merely a problem of social adaptation; it is a problem in which the foundational conditions for the realization of human subjecthood are being damaged.

"Humanity as a future-oriented subject" does not introduce a new risk-bearing subject, nor is it a new concept independent of the realization of subjecthood. It refers to the temporal and agential dimension of the realization of subjecthood: humans not only know, judge, and take responsibility in the present, but can also understand risks, revise institutions, organize action, and shape the future across generations. The fourth threshold is therefore a convergent criterion after the first three thresholds, testing whether local damage to capacities has already converged into damage to the overall future-directedness of humanity.

Together, these four thresholds restrict the scope of the expanded definition. They ensure that the paper does not upgrade all social impacts of AI into existential risks, and does not misread ordinary technological change as civilizational-level damage.

6.7 Risk Identification Comes before Governance Proposals

The expanded definition proposed in this paper serves risk identification first, rather than directly providing a governance proposal. AI safety, AI risk analysis, and AI governance often require concrete technical measures, institutional arrangements, responsibility frameworks, and policy designs. But before these tasks, there is a more fundamental question: what exactly are we trying to protect?

If "humanity" is understood only as a biological species, then governance goals will mainly concern the prevention of extinction, loss of control, permanent incapacitation, and closure of the future. If "humanity" also includes the layer in which humans realize subjecthood under civilizational conditions, then governance goals must also pay attention to whether the conditions for knowledge, judgment, responsibility, institutional participation, and meaning are being weakened. The task of risk identification is to provide a conceptual location that can be discussed, compared, and scrutinized before harm appears as catastrophe. Many AI risks are difficult to identify precisely because they do not appear as collapse but as convenience; not as coercion but as assistance; not as destruction but as substitution; not as disaster but as normalization.

6.8 Section Summary

This section has explained why AI makes biological survival, civilizational conditions, and category calibration unavoidable questions in existential-risk analysis. AI may threaten the biological survival of humanity through high-consequence technological systems; may weaken structures of human participation in the functional systems of civilization through deprivation by substitution; and may disturb the calibration mechanism for foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning through mediated reconfiguration. On this basis, the paper has proposed an expanded definition and four thresholds, avoiding both the upgrading of all social impacts of AI into existential risks and the reduction of existential risk to humanity to the single problem of species extinction.

The next section concludes the paper and explains the significance of this conceptual reconstruction for AI safety, AI risk analysis, and research on humanity's future.

7 | Conclusion: From Species Survival to the Realization of Subjecthood

This paper began with a prior question: when we say that AI may pose an existential risk to “humanity,” what exactly is the “humanity” at risk? The answer offered here is that the risk-bearing subject of AI-related existential risk to humanity is always humanity as an embodied species. Civilization, states, cultures, technological systems, the philosophical kernel, and AI systems are not risk-bearing subjects parallel to humanity. What this paper expands is not the risk-bearing subject itself, but the possible modes of existential harm to that subject.

To this end, the paper has undertaken a work of **conceptual engineering**. It distinguishes two basic layers of possible harm to the risk-bearing subject: the biological-survival layer and the civilizational-conditions layer. At the same time, within the civilizational-conditions layer, the paper separately identifies the category-calibration mechanism, namely the philosophical kernel. The biological-survival layer explains whether humanity continues to exist as an embodied species. The civilizational-conditions layer explains whether humans can still realize subjecthood through an intergenerational common world. The category-calibration mechanism explains whether foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning can still be clarified, maintained, and revised. The philosophical kernel is not a third parallel subject, but the category-calibration mechanism within the civilizational-conditions layer that maintains the operability of foundational categories.

The paper therefore provides neither a complete risk taxonomy nor a technical control scheme or policy checklist, but an **ontological coordinate system** needed at the level of risk identification in AI safety, AI risk analysis, and AI governance. Before discussing specific risk types, governance pathways, or technical solutions, we must first clarify the risk-bearing subject called “humanity” and the possible modes of harm to it. Only within this

coordinate system can technical research, policy discussion, philosophical analysis, and social observation be placed within a framework that is comparable, connectable, and open to critique.

Subsequent research can build upon this subject framework to further develop concrete taxonomies, mechanisms of emergence, and governance implications for AI-related existential risk to humanity. The contribution of this paper is to provide a clearer ontological starting point for that work: from species survival to the realization of subjecthood under conditions of human civilization.

Appendix A | Core Conceptual Glossary

This appendix gathers and defines the core concepts used in this paper. Its purpose is not to restate the argument of the main text, but to provide stable conceptual boundaries for the conceptual engineering carried out there. Each entry specifies, as far as necessary, the concept's definition, boundaries, and role in the paper's argument. Unless otherwise indicated, all concepts in this appendix serve the paper's analysis of the risk-bearing subject of AI-related existential risk to humanity.

I. Core Ontological Concepts

A.1 Conceptual Engineering

Conceptual engineering refers to the work of making explicit, evaluating, revising, and reconstructing the use, boundaries, implicit assumptions, and theoretical function of a concept. It does not merely interpret how a concept is ordinarily used in existing language. Rather, it asks whether the concept is sufficiently clear, whether it misleads, and whether it needs to be redrawn for a particular theoretical or practical context.

In this paper, the object of conceptual engineering is “humanity” as the risk-bearing subject of AI-related existential risk to humanity. The paper does not attempt to redefine all AI risks, nor does it directly propose a governance scheme. Instead, it asks: when we say that AI may threaten “humanity,” must humanity be understood only as a biological species, or must the analysis also take into account the foundational conditions that enable humanity, as an embodied species, to realize its subjecthood within civilization?

The conceptual-engineering function of this paper is to transform “humanity” from an unexamined background term into a core object of analysis for risk identification.

A.2 AI-Related Existential Risk to Humanity

AI-related existential risk to humanity refers to the risk, arising from the development, deployment, integration, or loss of control of AI systems, that AI may irreversibly destroy the human species; or render humanity permanently incapacitated, or otherwise permanently foreclose humanity's future; or undermine—irreversibly or in ways that are difficult to reverse—the foundational conditions under which humanity, as an embodied species, realizes its subjecthood within civilization.

This definition includes three modes of harm. First, humanity as a biological species may be irreversibly destroyed. Second, humanity may survive without becoming extinct, yet become permanently incapacitated or otherwise permanently lose an open future. Third, the human species and a nominal future may continue to exist, while the foundational conditions for the realization of human subjecthood under civilizational conditions are weakened irreversibly or in ways that are difficult to reverse.

These three modes of harm are not mutually exclusive categories, but a structure of coverage. Species extinction normally also entails the deprivation of humanity's future; permanent incapacitation is one core form of such deprivation; and the irreversible weakening of the conditions for subjecthood shows that even if humans continue to exist biologically, humanity may lose the conditions under which it can continue to exist as a future-oriented subject.

A.3 Risk-Bearing Subject

A **risk-bearing subject** is the entity that, within a given risk analysis, is regarded as liable to suffer fundamental, irreversible, or difficult-to-reverse harm. The concept answers the question: upon whom, or upon what, does the risk ultimately fall?

In this paper, the risk-bearing subject of AI-related existential risk to humanity is always humanity as an embodied species. Civilization, states, cultures, institutions, technological systems, the philosophical kernel, and AI systems may all function as structures through which risks arise, propagate, intensify, or become mediated. They are not, however, the risk-bearing subject in the sense used in this paper.

This concept prevents a common confusion: mistaking the system in which a risk occurs for the subject that bears the risk. AI may act upon economic systems, legal institutions, educational structures, knowledge systems, or category-calibration mechanisms. These effects become existentially significant only because they may damage the biological survival of humanity as an embodied species or the conditions under which humanity realizes its subjecthood.

A.4 Humanity

In this paper, **humanity** refers first to *Homo sapiens* as an embodied biological population. Humanity has bodily structures, nervous systems, capacities for perception, reproductive

mechanisms, physiological vulnerability, mortality, and dependence on ecological and material environments.

Humanity, however, does not exist only at the level of biological survival. Humanity as an embodied species also realizes subjecthood under civilizational conditions: through language, knowledge, institutions, structures of responsibility, public reason, historical memory, narratives of meaning, and the functional systems of civilization, humans acquire and sustain the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future.

Thus, “humanity” in this paper has two basic analytic layers: the biological-survival layer and the layer of subjecthood-realization under civilizational conditions. The former identifies humanity’s embodied basis of existence; the latter explains why humanity is not merely a biological population, but a historical form of existence capable of being a knower, judge, bearer of responsibility, participant in institutions, creator of meaning, and future-oriented subject.

A.5 Embodied Species

An **embodied species** is a biological population that cannot be reduced to a pure information structure, an abstract rational subject, or a computational pattern that can be arbitrarily migrated. In the case of humanity, embodiment includes bodies, nervous systems, perception, affect, vulnerability, mortality, relations of dependence, and intergenerational continuity.

Embodiment is not an external attribute added to human subjecthood. It is the biological precondition under which human civilization, responsibility, meaning, and future-directed action can be borne. Human judgment, learning, responsibility, and meaning-formation are tied to the body, finitude, temporality, and the capacity to live with the consequences of action.

In AI risk analysis, the concept of humanity as an embodied species resists a reductive tendency: the tendency to understand humans as cognitive patterns replaceable by more efficient information-processing systems. This paper argues that even if AI can perform many cognitive tasks, humans cannot be reduced to information-processing structures, because human existence as responsible agents, actors, and meaning-creators depends on embodied finitude.

A.6 Biological-Survival Layer

The **biological-survival layer** refers to the survival of humanity as *Homo sapiens*, an embodied biological population. It includes bodily structure, nervous systems, perceptual capacities, reproductive capacity, physiological vulnerability, mortality, and ecological dependence.

The biological-survival layer provides the physical bearer of human existence. If the human species becomes extinct, or if humanity permanently loses the capacity to reproduce,

act, or shape the future, then human civilization, structures of responsibility, narratives of meaning, and the realization of subjecthood all lose their bearer.

In this paper, the biological-survival layer corresponds to the baseline forms of traditional existential risk, including species extinction, permanent incapacitation, and future deprivation. The paper's expanded account of existential risk does not weaken the centrality of biological survival; rather, it builds upon that foundation in order to analyze modes of harm at the level of civilizational conditions.

A.7 Civilizational-Conditions Layer

The **civilizational-conditions layer** refers to the structural conditions upon which humanity, as an embodied species, depends in order to realize subjecthood within civilization. These conditions include language, knowledge systems, educational structures, legal institutions, mechanisms of responsibility, public reason, scientific communities, historical memory, technological practices, narratives of meaning, and the functional systems of civilization.

The civilizational-conditions layer is not a subject parallel to humanity, nor is it a bearer of risk independent of humanity. It is a constitutive condition for the realization of human subjecthood. If these conditions are weakened irreversibly or in ways that are difficult to reverse, the subject harmed is not “civilization itself” as an independent subject, but humanity as an embodied species in its capacity to realize subjecthood within civilization.

This concept shows that existential risk to humanity need not appear only as bodily destruction or species extinction. It may also appear as a condition in which humans remain alive, yet gradually lose the conditions for knowing what is true, forming judgment, bearing responsibility, participating in institutions, transmitting meaning, and acting toward the future.

A.8 Human Civilization

Human civilization is the intergenerational common world formed by humanity, as an embodied species, through language, knowledge, institutions, law, education, structures of responsibility, technological practices, functional systems of civilization, and narratives of meaning.

Human civilization is not the sum of population, is not identical with the state, is not merely culture, and cannot be reduced to a technological system. The human population provides the biological substrate; the state provides political and legal organization; culture provides meaning-expression and a lived interface; technology provides capacities for action. Human civilization is the common world formed through the intergenerational transmission, institutional organization, accumulation of knowledge, and generation of meaning among these elements.

This paper does not offer a general theory of civilization. Unless otherwise specified, “civilization” in the main text refers to human civilization. Human civilization is “human” not

because it contains no nonhuman technological components, but because its risk-bearing subject is humanity as an embodied species and because its normative focus is the realization of human subjecthood.

A.9 Modern Global Civilizational Ecology

The **modern global civilizational ecology** refers to the highly interconnected civilizational environment of contemporary human society, formed by multiple civilizational traditions, scientific systems, technological infrastructures, legal institutions, markets, states, international organizations, digital platforms, educational systems, media systems, and public reason.

The concept emphasizes two points. First, human civilization as a whole is not the universalization of one civilizational tradition, nor the enlargement of one state or culture. Second, modern global civilization has developed a number of shared interfaces, including scientific systems, international law, public-health systems, Internet protocols, financial networks, digital platforms, and AI infrastructures.

In this paper, the concept of modern global civilizational ecology explains why AI risks do not act only upon local states, cultures, or platforms. They may also act upon the knowledge, institutional, technological, and meaning-bearing interfaces upon which multiple civilizational traditions jointly depend.

II. Concepts of Civilizational Conditions and the Realization of Subjecthood

A.10 Constitutive Condition

A **constitutive condition** is a necessary condition under which a capacity, identity, or mode of existence can come into being. It is not an external or replaceable auxiliary tool.

Instrumental conditions help a subject perform a task; constitutive conditions participate in forming the subject's very capacity to perform that task. Language, for example, is not only a tool for expressing thought, but also a condition under which complex thought, exchange of reasons, and normative commitment can unfold. Similarly, education, structures of responsibility, public reason, functional systems of civilization, and narratives of meaning are not merely external supports for human subjecthood; they are constitutive conditions under which humans can exist as knowers, judges, bearers of responsibility, and creators of meaning.

This paper uses the concept to show that civilizational conditions are not decorative additions to human subjecthood, but part of the realization of human subjecthood itself.

A.11 Functional Systems of Civilization

Functional systems of civilization are the systems within human civilization that concretely bear knowledge production, institutional action, allocation of responsibility, educational transmission, economic organization, cultural expression, public governance, medical care, media communication, and the creation of meaning.

Typical functional systems of civilization include economic systems, state and governance systems, legal systems, educational systems, scientific and knowledge-production systems, cultural and media systems, medical systems, administrative systems, infrastructure systems, religious systems, and artistic systems.

Functional systems of civilization form an intermediate support layer between abstract civilizational conditions and concrete social mechanisms. Humans usually do not encounter “civilization” as an abstract whole. They enter civilizational conditions through schools, courts, laboratories, markets, media, government agencies, platform systems, families, religious traditions, artistic practices, and public services.

A.12 Structures of Human Participation

Structures of human participation are the ways in which humans participate in the operation, maintenance, and renewal of the common world as workers, learners, judges, creators, citizens, bearers of responsibility, and future-oriented subjects within the functional systems of civilization.

Human participation is not mere formal presence. It requires that humans be able to understand processes, form judgment, bear responsibility, create meaning, and exert substantive influence on institutional direction and the public future. If humans retain only formal confirmation within functional systems, while no longer genuinely understanding, judging, taking responsibility, creating, or shaping direction, then the realization of human subjecthood under civilizational conditions is weakened.

This concept provides the basis for “deprivation by substitution”: when AI makes the functional systems of civilization progressively less dependent on human participation, what is damaged is precisely the structure of human participation.

A.13 The Realization of Human Subjecthood within Civilization

The realization of human subjecthood within civilization refers to the process by which humanity, as an embodied species, actually acquires and sustains the capacities to know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future under civilizational conditions.

It is neither a merely biological attribute of the human species nor an independent attribute of civilization itself. It is the form in which human subjecthood is realized under civilizational conditions. It is precisely because humans can enter structures of knowledge, respon-

sibility, institutions, meaning, and future-directed action within civilization that humanity is not merely a biological population.

In this paper, the realization of subjecthood is the central object of risk at the civilizational-conditions layer. If AI systematically weakens these capacities and produces irreversible or difficult-to-reverse structural consequences, it may constitute an existential risk at the level of the realization of human subjecthood.

A.14 Capacities of Subjecthood

Capacities of subjecthood are the set of capacities actually exercised by humans under civilizational conditions as knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects.

This paper focuses on six capacities of subjecthood:

1. knowing what is true;
2. forming judgment;
3. bearing responsibility;
4. participating in and sustaining institutions;
5. transmitting and creating meaning;
6. acting toward the future.

These six capacities are not isolated from one another. Knowing what is true supports judgment; judgment supports responsibility; responsibility and institutional participation are mutually dependent; and meaning provides the justificatory structure for learning, creation, and commitment to the future. The systemic weakening of these capacities is the central object of this paper's expanded analysis of existential risk.

A.15 Foundational Civilizational Conditions

Foundational civilizational conditions are the minimal operating conditions required for humans to realize subjecthood within civilization. They are not the whole content of human civilization, but the foundational interfaces that allow humanity to continue existing as a future-oriented subject.

This paper focuses on five types of foundational civilizational conditions:

1. the calibrability of truth and evidence;
2. the attributability of responsibility and authorization;
3. the sustainability of subjecthood and judgment;
4. the openness of value-ordering to reflection;
5. the intelligibility and continuity of meaning-transmission.

These conditions are used to determine whether a risk at the civilizational-conditions layer touches the foundations of subjecthood-realization. If these conditions are systematically weakened, humans may still live in a highly functioning society, yet become increasingly unable to know what is true, form judgment, bear responsibility, participate in institutions, transmit meaning, and shape the future.

III. Concepts of Category Calibration

A.16 Foundational Categories

Foundational categories are the basic conceptual interfaces that the functional systems of civilization must share and invoke. They enable civilization to distinguish truth from falsehood, evidence from narrative, responsibility from exemption, subject from tool, value from metric, and meaning from arbitrary expression.

This paper focuses on six foundational categories:

1. truth;
2. evidence;
3. responsibility;
4. subjecthood;
5. value;
6. meaning.

Foundational categories are not theoretical terms monopolized by any one civilizational tradition. They are key operational interfaces upon which the modern global civilizational ecology depends. Different traditions may understand these categories differently, but insofar as a civilizational system requires knowledge, institutions, responsibility, action, and meaning, it must invoke these categories in some form.

A.17 Philosophical Kernel

The **philosophical kernel** is the category-calibration mechanism, internal to the civilizational-conditions layer, that maintains the operability of foundational categories. Because what it calibrates are the foundational categories jointly invoked by the various functional systems of civilization, rather than the local rules internal to any single functional system, it may also be described as a category-calibration meta-mechanism. Specifically, it refers to the mechanism through which foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning are made explicit, tested, clarified, maintained, and revised.

The philosophical kernel is not academic philosophy as a discipline, not an independent entity within civilization, and not an independent risk-bearing subject. Nor is it a module that can be separately installed, replaced, or restarted. It is called a “kernel” because it functionally performs the calibration of foundational category interfaces.

In this paper, the philosophical kernel explains how the functional systems of civilization can continue to operate while losing their intelligibility, accountability, and capacity for correction when their underlying categories become miscalibrated.

A.18 Category-Calibration Mechanism

The **category-calibration mechanism** is the historical process through which foundational categories are made explicit, tested, criticized, maintained, and revised. It is not the function of any single institution, but a continuing process distributed across science, law, education, public discourse, institutional reform, cultural practice, and historical experience.

The category-calibration mechanism concerns questions such as: what counts as truth, what counts as evidence, who can bear responsibility, what constitutes a subject, what is worth optimizing, and what remains meaningful.

When the category-calibration mechanism fails, the real-world manifestation is that systems such as science, law, education, governance, media, and culture may continue operating, yet become increasingly unable to stably distinguish truth from fabrication, evidence from narrative, responsibility from exemption, subject judgment from formal confirmation, value goals from optimization metrics, and meaning-creation from content generation.

A.19 Category of Subjecthood

The **category of subjecthood** is the conceptual structure concerning who can judge, consent, authorize, act, bear responsibility, and represent themselves. It is one of the foundational categories and serves functional systems of civilization such as law, politics, education, ethics, and public governance.

The category of subjecthood is distinct from the realization of subjecthood. The category of subjecthood is the conceptual interface through which the capacities of subjects are understood and institutionalized; the realization of subjecthood is the process by which humans actually exercise capacities such as knowing what is true, forming judgment, bearing responsibility, participating in institutions, creating meaning, and shaping the future.

This distinction avoids circularity. One condition for the realization of subjecthood is the clarity of the category of subjecthood; the claim is not the tautology that “subjecthood depends on subjecthood.”

A.20 Category Network

The **category network** is the mutually supporting relation among foundational categories such as truth, evidence, responsibility, subjecthood, value, and meaning.

Truth requires evidence for calibration; responsibility requires subjecthood as its bearer; subjecthood requires judgment for its realization; value requires meaning for interpretation; and meaning requires truth, responsibility, and value if it is not to collapse into arbitrary narrative. The miscalibration of any one foundational category may affect the understanding and operation of the others.

The concept of the category network explains why AI's impact on foundational categories is often not a single-point disturbance, but may instead appear as a linked drift among multiple category interfaces.

A.21 Systemic Drift of the Category Network

Systemic drift of the category network refers to the linked miscalibration among truth, evidence, responsibility, subjecthood, value, and meaning when AI or other structural forces simultaneously alter the practical conditions under which multiple category interfaces operate.

In such a condition, the functional systems of civilization may continue operating, and may even operate more efficiently, but their operation increasingly depends on automation, prediction, ranking, and optimization rather than understanding, judgment, responsibility, and meaning.

This concept describes the deeper consequence of mediated reconfiguration: the risk is not merely that a given system malfunctions, but that the way the functional systems of civilization invoke foundational categories undergoes systemic drift.

A.22 The Mapping Relation among Foundational Categories, Foundational Civilizational Conditions, and Capacities of Subjecthood

The **mapping relation among foundational categories, foundational civilizational conditions, and capacities of subjecthood** refers to the functional connection among three levels in this paper:

1. foundational categories provide conceptual interfaces;
2. foundational civilizational conditions provide operating conditions;
3. capacities of subjecthood are the capacities actually exercised by humans within civilization.

This mapping relation can be summarized as follows: foundational categories support foundational civilizational conditions, and foundational civilizational conditions support capacities of subjecthood.

The concept prevents the three lists from drifting apart. It shows that the paper is not arbitrarily listing abstract concepts, but constructing a layered relation from conceptual interfaces to civilizational operating conditions and then to actual human capacities.

IV. Concepts of AI Risk Mechanisms and Thresholds

A.23 AI as a Civilizational Mediator

AI as a civilizational mediator means that AI is not merely an external tool, but may enter the structures of knowledge, evidence, responsibility, judgment, institutions, and meaning within human civilization, becoming a mediating force that affects the conditions for the realization of human subjecthood.

AI may become a tool, platform, infrastructure, or mediating structure within human civilization, but it does not thereby become the risk-bearing subject in this paper. The risk significance of AI lies not only in the possibility that it may cause external disasters, but also in the possibility that it may change how humans know what is true, form judgment, bear responsibility, participate in institutions, and create meaning.

A.24 Deprivation by Substitution

Deprivation by substitution refers to the process by which AI, as a more efficient, more replicable, and more scalable substitute, progressively takes over human roles in labor, cognition, creation, governance, and judgment within the functional systems of civilization, thereby marginalizing humans within, or displacing them from, the key functional systems of the common world.

Deprivation by substitution primarily weakens the position of humans as participants in, and shapers of the direction of, the common world. Its typical form is not the immediate disappearance of humans, but a condition in which humans still consume, confirm, vote, or receive services, while exercising decreasing substantive influence over resource allocation, institutional direction, knowledge production, and cultural narratives.

This concept corresponds to damage to the structures of human participation at the civilizational-conditions layer.

A.25 Mediated Reconfiguration

Mediated reconfiguration refers to the condition in which humans remain within the functional systems of civilization and retain forms of choice, approval, signature, learning, creation, and participation, while the preconditions of these activities have become deeply mediated by AI.

In mediated reconfiguration, humans still receive information, but the points of entry, summary structures, and interpretive frames of that information may be pre-organized by systems; humans still make decisions, but the visible options, risk rankings, and generated reasons may be shaped by AI; humans still bear responsibility, but the process by which judgment is formed becomes increasingly difficult to understand and trace; humans still learn and create, but learning processes, standards of creation, and evaluations of meaning increasingly depend on system output.

Mediated reconfiguration primarily weakens human understanding, judgment, responsibility, and meaning-creation within the process of participation, and may further disturb the mechanism of category calibration.

A.26 Irreversible or Difficult to Reverse

Irreversible or difficult to reverse is the key temporal standard used in this paper to distinguish existential risk from ordinary major risk.

At the biological-survival layer, irreversibility primarily refers to absolute physical or biological nonrecoverability, such as species extinction, rupture of the reproductive chain, or permanent incapacitation.

At the civilizational-conditions layer, irreversibility or difficulty of reversal primarily refers to historical, institutional, and cognitive forms of nonrecoverability, such as institutional path dependence, breakdowns in intergenerational capacity formation, erosion of key judgment capacities, and the failure of a society to recognize the loss of those capacities as a loss.

This paper does not quantify “difficult to reverse” by a fixed duration or a single metric. It is a standard of conceptual identification whose concrete operationalization requires further empirical and governance research.

A.27 Humanity as a Future-Oriented Subject

Humanity as a future-oriented subject means that humanity does not merely enter the future passively, but is capable, across generations, of understanding risks, forming judgment, bearing responsibility, organizing action, revising institutions, and shaping the future.

It is not a new risk-bearing subject, nor a concept independent of the realization of subjecthood. It is the temporal and agential dimension of the realization of subjecthood. In other words, humanity as a future-oriented subject means that humanity can direct knowledge, judgment, responsibility, institutional participation, and meaning-creation toward the future, and thereby protect or renew the common world.

This concept is used in the fourth existential-risk threshold: a risk has existential-risk significance only if it causes the harms identified by the first three thresholds to converge into overall damage to humanity’s capacity as a future-oriented subject.

A.28 Existential-Risk Thresholds

Existential-risk thresholds are the four standards used in this paper to determine whether a risk to the realization of human subjecthood at the civilizational-conditions layer reaches the level of existential risk.

For a risk to enter the scope of existential risk at the level of the realization of human subjecthood, it must at least:

1. act upon foundational civilizational conditions rather than local functions;
2. have cross-domain diffusion capacity rather than being an isolated accident or single misuse;
3. be capable of causing irreversible or difficult-to-reverse structural damage;
4. cause the harms identified in the first three thresholds to converge into overall, inter-generational damage to humanity's capacity as a future-oriented subject.

These thresholds primarily apply to risks to the realization of subjecthood at the civilizational-conditions layer. Species extinction, permanent incapacitation, or the permanent foreclosure of humanity's future directly constitutes existential risk by its own nature and need not be assessed through these four thresholds.

A.29 Ordinary Social Risk

Ordinary social risk refers to risk that affects a local social function, institutional process, industry structure, or technological application. Such risk may be very serious, but it usually has a relatively determinate governance object and path of repair.

For example, the replacement of a profession, the restructuring of an industry, the automation of an institutional process, or errors in an information system may all constitute serious social risks, but they do not necessarily constitute existential risks to humanity.

The difference between ordinary social risk and risk to the realization of human subjecthood lies not in severity alone, but in whether the risk acts upon the foundational conditions under which humanity can continue to exist as a future-oriented subject.

A.30 Risk to the Realization of Human Subjecthood

Risk to the realization of human subjecthood refers to risk that acts upon the foundational conditions under which humans realize capacities for knowledge, judgment, responsibility, institutional participation, meaning-creation, and future-directed action under civilizational conditions.

It differs from ordinary social risk. Ordinary social risk usually acts upon local functions; risk to the realization of human subjecthood acts upon more foundational civilizational conditions and may affect whether humans can continue to exist as knowers, judges, bearers of responsibility, institutional participants, creators of meaning, and future-oriented subjects.

When such a risk acts upon foundational civilizational conditions, has cross-domain diffusion capacity, is capable of causing irreversible or difficult-to-reverse structural damage, and causes these harms to converge into overall, intergenerational damage to humanity's capacity as a future-oriented subject, it may enter the scope of AI-related existential risk to humanity.

A.31 Paradigm Extension

Paradigm extension refers to expanding the range of objects or modes of harm recognized by risk identification while preserving the core standing of existing existential-risk frameworks.

The paradigm extension proposed in this paper is not a paradigm replacement. It does not deny species extinction, permanent incapacitation, and future deprivation as baseline forms of existential risk. Rather, it further argues that AI may also constitute existential risk to humanity by undermining, irreversibly or in ways that are difficult to reverse, the foundational conditions under which humanity realizes subjecthood within civilization.

A.32 Risk-Identification Framework

A **risk-identification framework** is an analytic framework used before specific risk classification, governance design, or technical control. It determines the object of risk, the modes of harm, the thresholds of judgment, and the boundaries of relevant concepts.

This paper provides a risk-identification framework, not a complete taxonomy of risks and not a technical governance proposal. It answers a question prior to all of these: what is AI risk analysis trying to protect? The answer of this paper is that it must protect not only the survival of humanity as a biological species, but also the foundational conditions under which humanity can continue to realize subjecthood within civilization.

A.33 The AGI Era

The **AGI era**, as used in this paper, does not mean that AGI in the strict sense has already been achieved, nor does it depend on the claim that a universally accepted AGI threshold has been crossed.

It refers to an unfolding historical transition: AI systems are becoming more general, more autonomous, more capable of action, and increasingly embedded in knowledge production, institutional processes, judgment formation, technological research and development, and meaning generation.

This concept marks a historical condition: even before genuine AGI appears, AI has already begun to change the civilizational conditions under which human subjecthood is realized.

A.34 Ontological Coordinate System

An **ontological coordinate system** is the risk-identification framework this paper provides for AI-related existential risk to humanity. By clarifying the risk-bearing subject, the layers of possible harm, the category-calibration mechanism, the functional systems of civilization, and the relevant risk thresholds, it gives different AI risk discussions a structure in which they can be compared, connected, and critically examined.

In this paper, the ontological coordinate system includes:

1. one risk-bearing subject: humanity as an embodied species;
2. two basic layers of possible harm: the biological-survival layer and the civilizational-conditions layer;
3. one internal mechanism: the category-calibration mechanism;
4. one intermediate support layer: functional systems of civilization and structures of human participation;
5. two AI mechanisms of action: deprivation by substitution and mediated reconfiguration;
6. a set of existential-risk thresholds.

This concept clarifies the relation between this paper and risk taxonomies, technical governance, and policy proposals. The paper does not directly replace those forms of work; it provides the prior conceptual coordinates within which they can be located.

Notes

[Note 1]

The “AGI era” in this paper does not presuppose that AGI in the strict sense has already been achieved, nor does it depend on the crossing of any generally accepted AGI threshold. It refers to an ongoing historical transition: AI systems are becoming more general, more autonomous, more capable of action, and increasingly deeply embedded in knowledge production, institutional processes, judgment formation, technological research and development, and meaning generation. Several risk mechanisms discussed in this paper may already appear, before the arrival of genuine AGI, in the form of advanced AI, generative AI, agentic AI, and automated systems.

[Note 2]

This paper treats AI as a civilizational mediator that may reorganize structures of human knowledge, action, responsibility, and meaning. This problem-consciousness is adjacent to the tradition of technological mediation in philosophy of technology and postphenomenology. For relevant background, see Heidegger (1977), Arendt (1958), Jonas (1984), Win-

ner (1980), and Verbeek (2011). Pezzano and Pavanini (2025) have recently further distinguished technological support, technological mediation, and technological constitution; this can serve as a contemporary theoretical background for understanding how technology may move from external support toward mediation and even toward constitutive conditions. This paper does not claim that technology has a single deterministic direction, but focuses on how AI, as a technological mediator, may change the civilizational conditions under which human subjecthood is realized.

[Note 3]

Existing discussions of existential risk and AI risk usually emphasize species extinction, permanent incapacitation, future deprivation, loss of control, the control problem, and the irreversible loss of long-term future potential. See Bostrom (2013, 2014), Ord (2020), Russell (2019), and Carlsmith (2022). Recent philosophical and legal scholarship has also begun to systematize AI existential risk. Cappelen, Goldstein, and Hawthorne (2025) propose a taxonomy of AI existential risk from the perspective of AI survival stories; Tokson and Arbel (2026), from the perspective of legal governance, distinguish human-directed risks, accident risks, and loss-of-control risks, and discuss how legal institutions may identify and respond to these risks under uncertainty. This paper is adjacent to these studies, but does not propose a new risk taxonomy or regulatory scheme. Rather, it more fundamentally clarifies the “humanity” that is the risk-bearing subject in these risks and the possible modes of harm to it.

[Note 4]

For the conceptual-engineering orientation of this paper, see Cappelen (2018), Chalmers (2025), and Haslanger (2000) on conceptual revision, conceptual design, and normative conceptual analysis. This paper borrows the problem-consciousness of conceptual engineering, but its specific object is not a debate internal to philosophy of language. It is the risk-bearing subject in AI risk theory: when we say that AI threatens “humanity,” what exactly is the “humanity” in question? This paper does not redefine all AI risks; it clarifies the conceptual boundaries of “humanity” as the risk-bearing subject in AI-related existential risk to humanity.

[Note 5]

The “hardware—kernel—distribution—application layer” analogy is used only as a functional explanation, not as an ontological model. It does not imply that human civilization can be modularly decomposed, replaced, reinstalled, or migrated like a computer system; nor does it imply that the philosophical kernel is an entity-module that can be separately installed, replaced, or restarted. The analogy is used only to explain four functional relations: the human species provides the biological substrate; the philosophical kernel calibrates foundational category interfaces; human civilization provides the complete operating environment; and the functional systems of civilization provide the application-layer interfaces

through which humans actually enter the common world.

[Note 6]

This paper does not claim that AI will necessarily weaken the conditions for the realization of human subjecthood. AI may also strengthen education, scientific research, public services, medical diagnosis, accessibility support, knowledge access, institutional transparency, and public collaboration. The paper focuses on another possibility: only when the ways in which AI is integrated systematically weaken the conditions under which humans know what is true, form judgment, bear responsibility, participate in and sustain institutions, transmit and create meaning, and act toward the future, and when these effects have cross-domain, structural, and difficult-to-reverse consequences, do they enter the scope of existential risk at the level of the realization of human subjecthood as discussed here.

[Note 7]

For the possibility that AI risk may arise through the gradual weakening of human influence over large-scale social systems such as the economy, culture, and the state, see Kulveit et al. (2025), “Gradual Disempowerment: Systemic Existential Risks from Incremental AI Development.” In the coordinate system of this paper, such research can be situated within the civilizational-conditions layer, especially in relation to the functional systems of civilization and structures of human participation. It is especially close to what this paper calls deprivation by substitution: AI gradually replaces humans in labor, cognition, cultural production, governance participation, and social coordination, thereby weakening humans’ substantive influence over the common world. This paper absorbs that risk intuition, but does not take it as its main framework.

[Note 8]

On the relation between generative AI use, critical thinking, cognitive effort, and modes of task supervision, see Lee et al. (2025), a survey study of knowledge workers. On students’ over-reliance on AI dialogue systems and its effects on cognitive abilities, see the systematic review by Zhai, Wibowo, and Li (2024). These studies do not directly prove existential risk, but they provide empirical background for what this paper calls mediated reconfiguration: when generative AI or AI dialogue systems are embedded in knowledge work and learning processes, practices of critical thinking, cognitive effort, burdens of judgment, and modes of task supervision may be redistributed.

[Note 9]

On the possibility that AI may generate deep risks by weakening human autonomy, decision-making capacities, and skill practices, see Prunkl (2022) on human autonomy in the age of AI, and Krook (2025) on autonomy decline and skill degradation as existential risk. In the

coordinate system of this paper, such discussions can be situated within the problem of weakened capacities of subjecthood at the civilizational-conditions layer, and overlap with both deprivation by substitution and mediated reconfiguration. The main axis of this paper is not autonomy theory itself, but the risk-bearing subject and the conditions under which its subjecthood is realized.

[Note 10]

The discussion in this paper of truth, evidence, knowledge communities, public judgment, and information ecology is connected to the tradition of social epistemology. For relevant background, see Goldman (1999), Longino (1990), Fricker (2007), O'Connor and Weatherall (2019), and Nguyen (2020). More directly, Ortmann (2025) discusses epistemic dependence on AI in science in *Synthese*, and argues that the opacity of AI tools does not necessarily introduce novel problems for social epistemology. This paper is adjacent to that discussion, but operates at a different level. It does not ask whether AI opacity constitutes a new problem for social epistemology, but whether, when AI deeply enters the functional systems of civilization and changes category interfaces such as truth, evidence, responsibility, subjecthood, value, and meaning, this mediated reconfiguration may touch the foundational conditions for the realization of human subjecthood.

[Note 11]

On whether large language models can be understood as part of the extended mind, see Smart, Clowes, and Clark (2025), a discussion of LLMs and the extended mind in *Synthese*. That paper examines LLMs within the extended-mind framework and addresses issues such as external cognitive resources, agency, and cognitive control. This paper is adjacent to that discussion, but does not center on whether LLMs constitute the extended mind. It focuses on whether, when AI systems organize points of information entry, reason generation, option ranking, and judgment environments, humans still retain the substantive conditions for forming judgment, bearing responsibility, and realizing subjecthood.

[Note 12]

“Irreversible or difficult to reverse,” as used in this paper, refers at the civilizational-conditions layer primarily to difficulty of recovery in historical, institutional, and cognitive senses, rather than to a single form of physical irreversibility. Relevant discussions may be connected to research on path dependence, institutional persistence, and capacity formation, such as Arthur (1989) on technological lock-in and increasing returns, Pierson (2000) on path dependence in politics, North (1990) on institutional change, and Cunha and Heckman (2007) on the technology of skill formation. This paper does not quantify “difficult to reverse” in terms of a fixed duration or a single indicator, but treats it as a conceptual standard in risk identification.

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